

Beta-Blocker (Bisoprolol) vs Calcium-Channel Blocker (Verapamil) in Nonobstructive Hypertrophic Cardiomyopathy



A Randomized Triple-Crossover Physiologic Trial

Louise Bjerregaard, MD, PhD,^{a,b} Anne M. Dybro, MD, PhD,^a Lotte Saaby, MD, PhD,^c Steen Hvitfeldt Poulsen, MD, PhD, DMSci,^{a,b} Torsten Bloch Rasmussen, MD, PhD,^a Morten Steen Kvistholm Jensen, MD, PhD^{a,b}

ABSTRACT

BACKGROUND Randomized evidence supporting treatment with beta-blockers and calcium antagonists in nonobstructive hypertrophic cardiomyopathy (HCM) is absent.

OBJECTIVES The authors sought to analyze the effect of bisoprolol and verapamil treatment in nonobstructive HCM in a randomized, double-blinded, placebo-controlled triple-crossover trial.

METHODS Nonobstructive HCM patients with ≥ 1 marker of disease severity (NYHA functional class $\geq \text{II}$, N-terminal pro-B-type natriuretic peptide [NT-proBNP] > 300 ng/L, or documented nonsustained ventricular tachycardia) were included. Each patient underwent 3 treatment periods with target doses of 7.5 mg bisoprolol, 360 mg verapamil, and placebo. Outcomes were evaluated after 2 weeks on steady-state treatment. The primary outcome was peak oxygen consumption (pVO_2). In addition, exercise response, symptom, biomarker, structural, and myocardial changes were evaluated.

RESULTS Thirty-two patients (34% women) aged 54 ± 15 years were included. The pVO_2 was 25.7 ± 8.7 mL/kg/min with bisoprolol, 28.2 ± 8.6 mL/kg/min with verapamil and 28.7 ± 8.7 mL/kg/min with placebo. The adjusted mean difference in pVO_2 was -1.8 mL/kg/min ($P = 0.013$) bisoprolol vs verapamil, -2.5 mL/kg/min ($P = 0.002$) bisoprolol vs placebo and -0.7 mL/kg/min ($P = 0.990$) verapamil vs placebo. The peak heart rate was lower with bisoprolol (-37 beats/min; $P < 0.001$) and with verapamil (-17 beats/min; $P < 0.001$) compared with placebo. Neither bisoprolol nor verapamil changed the oxygen consumption at anaerobic threshold or the minute ventilation/ CO_2 production slope. Global longitudinal strain was improved by verapamil (-1.1% ; $P = 0.001$) but unchanged by bisoprolol (-0.6% ; $P = 0.263$) compared with placebo. Compared with placebo, treatment with bisoprolol reduced Kansas City Cardiomyopathy Questionnaire Overall Summary Score (KCCQ-OSS; -6.6 points; $P = 0.001$), increased NT-proBNP ($+165$ ng/L; $P = 0.006$), increased left atrial volume index (LAVI; $+13.0$ mL/m²; $P < 0.001$), and increased tricuspid regurgitation (TR) pressure gradient ($+4.3$ mm Hg; $P = 0.049$). Treatment with verapamil did not change KCCQ-OSS (-2.3 points; $P = 0.668$), LAVI ($+5.2$ mL/m²; $P = 0.194$), or TR pressure gradients ($+1.1$ mm Hg; $P = 1.000$), but reduced NT-proBNP (-177 ng/L; $P < 0.001$) compared with placebo. NYHA functional classification was unchanged by treatments.

CONCLUSIONS In nonobstructive HCM, treatment with bisoprolol reduced pVO_2 , whereas it was unchanged with verapamil. There were a number of significant differences in exercise response, structural changes, myocardial function, symptoms, and biomarkers. This study provides new insights into the mechanisms of these drugs that might be taken into consideration in management of nonobstructive HCM. (JACC. 2026;87:1063-1083) © 2026 by the American College of Cardiology Foundation.

From the ^aDepartment of Cardiology, Aarhus University Hospital, Aarhus, Denmark; ^bDepartment of Clinical Medicine, Aarhus University, Aarhus, Denmark; and the ^cDepartment of Cardiology, Odense University Hospital, Odense, Denmark. The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

Manuscript received August 1, 2025; revised manuscript received November 13, 2025, accepted November 14, 2025.

ABBREVIATIONS AND ACRONYMS

AT	= anaerobic threshold
BB	= beta-1 receptor blocker
CCB	= calcium-channel blocker
CPET	= cardiopulmonary exercise testing
GLS	= global longitudinal strain
HCM	= hypertrophic cardiomyopathy
HFpEF	= heart failure with preserved ejection fraction
hs-cTnl	= high-sensitivity cardiac troponin I
LAVI	= left atrial volume index
LV	= left ventricular
LVOT	= left ventricular outflow tract
NT-proBNP	= N-terminal pro-B-type natriuretic peptide
pVO₂	= peak oxygen consumption
RPP	= rate-pressure product
TR	= tricuspid regurgitation
VCO₂	= CO ₂ production
VE	= minute ventilation
VO₂	= oxygen consumption

Hypertrophic cardiomyopathy (HCM) is associated with limiting heart failure symptoms resulting in impaired functional capacity.^{1–3} For decades, first-line therapy for HCM has been cardioselective beta-1 receptor blockers (BBs) or nondihydropyridine calcium-channel blockers (CCBs).^{4,5} These recommendations are based on small clinical trials,^{6–12} registries,¹³ clinical experience, and expert opinions.¹⁴ The majority of studies assessed the effect of treatment in patients with obstructive HCM^{15–17} or heterogeneous groups including both obstructive and non-obstructive HCM.^{18,19} No randomized trials have assessed the efficacy of BBs or CCBs in an exclusively nonobstructive HCM cohort. There is growing evidence for medical treatment in obstructive HCM,^{20,21} yet the clinical and hemodynamic phenotypes of obstructive and nonobstructive HCM are markedly different. Thus, extrapolation effects of medical treatment from obstructive to nonobstructive HCM may not be appropriate.

SEE PAGE 1084

The aim of the present trial was to provide a comprehensive and contemporary evaluation of functional capacity, exercise response, myocardial function, structural changes, symptoms biomarker response of treatment with bisoprolol and verapamil in patients with nonobstructive HCM in a randomized, double-blind, placebo-controlled, triple-crossover trial.

METHODS

The trial was conducted at the Department of Cardiology, Aarhus University Hospital, and the Department of Cardiology, Odense University Hospital, Denmark. It adhered to the principles outlined in the Declaration of Helsinki and received approval from the local ethics committee and the Danish medicines agency (EudraCT no. 2021-006953-77). Subsequently, it was transferred to the Clinical Trials Information System under the European Medicines Agency (identifier: 2024-515345-17-00). The trial was registered on ClinicalTrials.gov (NCT05569382). Written informed consent was obtained from every participant, and the study was externally monitored by the Danish Good Clinical Practice units. Further details on monitoring are provided in the [Supplemental Methods](#).

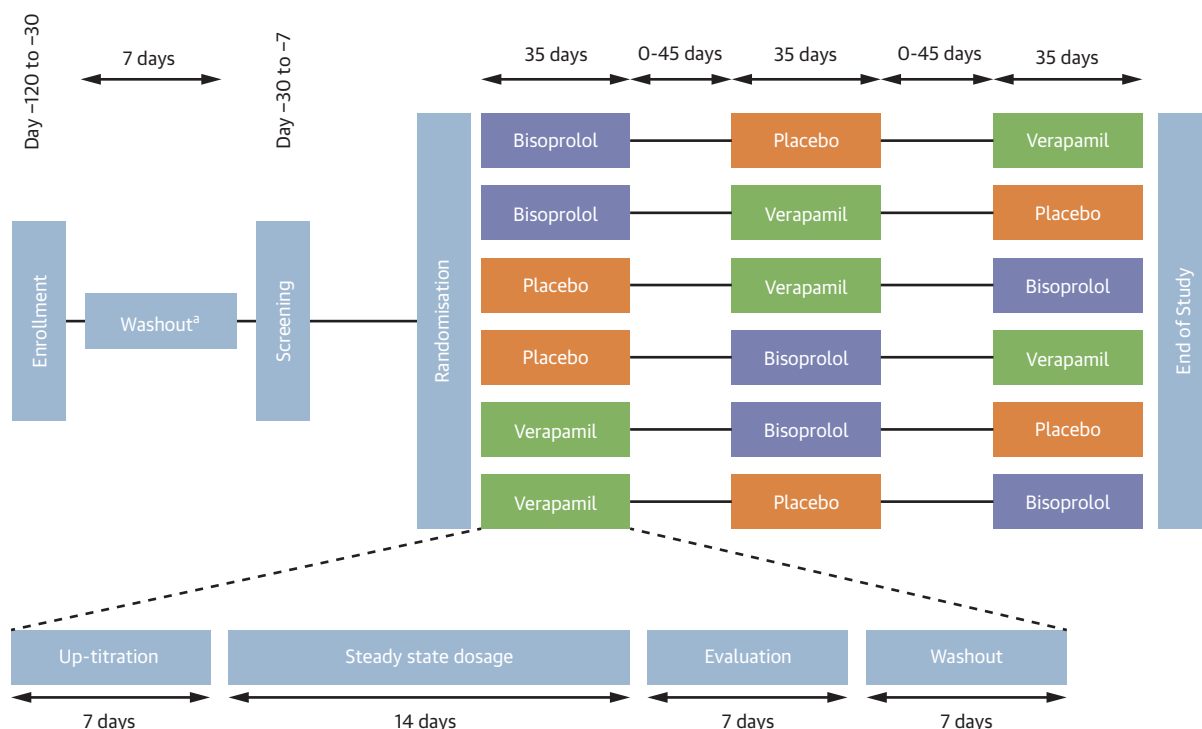
STUDY DESIGN. We conducted a dual-center, randomized, double-blind, placebo-controlled, triple-crossover trial to assess the treatment effects of bisoprolol and verapamil in patients with non-obstructive HCM. The trial was organized into 3 stages. The data presented here are from stage 1 of the trial; stages 2 and 3 are ongoing ([Supplemental Figure 1](#), [Supplemental Table 1](#)). In this crossover design, each patient received all 3 treatments (bisoprolol, verapamil, and placebo) in a randomized sequence of 3 treatment periods ([Figure 1](#)).

PATIENTS. Adult patients (aged ≥ 18 years) with nonobstructive HCM were screened for eligibility. Nonobstructive HCM was defined as left ventricular (LV) wall thickness ≥ 15 mm and a lifetime absence of significant left ventricular outflow tract (LVOT) obstruction >30 mm Hg at rest or during any provocation. Furthermore, patients had to present with at least 1 marker of disease severity: NYHA functional class $\geq \text{II}$ (or a history of this before treatment with BB or CCB), elevated N-terminal pro-B-type natriuretic peptide (NT-proBNP) >300 ng/L, or nonsustained ventricular tachycardia, defined as ≥ 3 consecutive ventricular beats at rate >120 beats/min within the last 2 years before screening. Additional inclusion and exclusion criteria are listed in the [Supplemental Methods](#) and [Supplemental Table 2](#).

STUDY DRUGS. The study drugs, provided by the hospital pharmacy, were uniformly encapsulated to ensure identical appearance across treatment periods. Each capsule contained 2.5 mg bisoprolol, 120 mg verapamil, or placebo. Target doses were 3 capsules daily, equal to 7.5 mg bisoprolol, 360 mg verapamil, or placebo. The treatment sequence was exclusively known by the hospital pharmacy, and the patients, investigators, and site staff remained blinded until external monitoring was completed and all outcomes were locked in an electronic database.

ENROLLMENT AND SCREENING. Patients were identified at outpatient clinics. After written informed consent, patients receiving BB or CCB treatment at the time of enrollment underwent a 1-week washout period ensuring at least 3 days without treatment before the screening visit. This procedure ensured a minimum of 27 days from latest intake of usual BB or CCB treatment before outcome evaluation in the first treatment period. The remaining standard-care medications were unchanged throughout the study period. Information on symptoms, previous non-sustained ventricular tachycardia and LVOT obstruction was evaluated from patient records.

FIGURE 1 Study Design



A randomized, double-blind, placebo-controlled, triple cross-over trial. Each 35-day treatment period included 1 week of dose up-titration, 2 weeks of stable dose, 1 week of outcome evaluation, and 1 week wash-out. ^aWashout of usual treatment with beta-blocker or verapamil

At screening, resting transthoracic echocardiography was performed according to standard guidelines, with assessment of LVOT obstruction at rest, during Valsalva maneuver, in a standing position, and standing after 1 minute of squatting. In addition, symptom burden and health status evaluations were assessed, together with biomarker testing and cardiovascular examinations such as 12-lead electrocardiography, spirometry, and vital signs.

RANDOMIZATION, BLINDING, AND INTERVENTION.

Patients were randomized to 1 of 6 possible treatment sequences (Figure 1), comprising different orders of bisoprolol, verapamil, and placebo, by means of a computer-based system managed by the pharmacy at Aarhus University Hospital. Randomization was conducted in blocks of 6 patients each, with 1:1:1:1:1:1 allocation to the 6 sequences within every block, thereby ensuring balanced allocation. Each center had separate randomization blocks.

Each treatment period spanned 35 days, including 1 week (days 1-7) up-titration, during which the study drug was increased from 1 capsule to the target dose of 3 capsules or maximum tolerated dosage with no or only tolerable side-effects. Adverse events and

tolerability were assessed through telephone interviews after the standardized 1 week of up-titration, and the maximal tolerable dosage was determined accordingly. Patients continued the stable dose for 2 weeks (days 8-21) before the evaluation of outcomes. If further dose adjustments were necessary, this was permitted; however, to ensure steady state at the time of evaluation, the dosing was maintained stable at least 1 week before evaluation of outcomes. After the evaluation, the dosing remained stable for 1 week (days 22-28) of Holter monitoring (part of stage 3), before undergoing individualized 1 week wash-out (days 29-35) (Figure 1). This treatment routine was repeated 3 times for each patient, with 0-45 days from the end of 1 treatment period to the beginning of the next period. Consequently, and according to the study protocol, there was a minimum of 17 and maximum of 66 days from the last dose of the previous drug to the evaluating the effect of the next drug in the following treatment period, allowing for sufficient washout. Compliance with the study drug was assessed through capsule counts on the day of evaluation.

OUTCOMES. Primary outcome. The primary outcome was difference in peak oxygen consumption (pVO_2) obtained from cardiopulmonary exercise testing (CPET) between bisoprolol and verapamil.

Predefined secondary and exploratory outcomes. When evaluating exercise response, the predefined secondary outcomes included comparison of percentage of predicted pVO_2 , the ventilatory efficiency according to minute ventilation (VE)/ CO_2 production (VCO_2) slope, and oxygen consumption (VO_2) at anaerobic threshold (AT). Explorative outcomes included evaluation of heart rate response, blood pressure, resting VO_2 , rate pressure product (RPP), and O_2 -pulse.

LV systolic function was evaluated with LV ejection fraction, global longitudinal strain (GLS), and ventricular volumes. Evaluation of diastolic function included peak E- and A-wave velocities, the E/A ratio, peak E deceleration time, mitral annular tissue e' velocities, E/ e' ratio, peak tricuspid regurgitation (TR) pressure gradients and left atrial volume index (LAVI). Symptom burden was assessed by NYHA functional class and the Kansas City Cardiomyopathy Questionnaire (KCCQ), and biomarkers by NT-proBNP and high-sensitivity cardiac troponin I (hs-cTnI). Predefined secondary and explorative outcomes were analyzed in comparison between bisoprolol and verapamil, as well as bisoprolol vs placebo and verapamil vs placebo. For details, see the [Supplemental Methods](#).

EVALUATION OF OUTCOMES. Cardiopulmonary exercise testing. To determine the primary outcome, pVO_2 , gas exchanges were obtained using breath-by-breath technique during a standardized ramp protocol performed on an upright bicycle (Ergoline VIAsprint 150p or Ebike Comfort Ergometers; General Electric). The work rate was tailored to each patient's physical condition as judged by the investigator, targeting for peak exercise to be reached after approximately 8 to 12 minutes of exercise. The ramp protocol started with a 3-minute cycling phase at zero resistance to allow for adaptation before the workload was applied at either 10 W + 10 W/min, 25 W + 12 W/min, or 25 W + 20 W/min. The recovery phase was conducted at zero resistance for a minimum of 3 minutes. Heart rate monitoring was continued until heart rate dropped below 100 beats/min or until a maximum duration of 10 minutes was reached. Patients were encouraged to exercise until exhaustion while maintaining a cadence of 60 to 65 rounds per minute. Continuous 12-lead electrocardiography and pulse oximetry, along with blood pressure every 2 minutes, were obtained. The same exercise protocol and technical equipment was

consistently applied across all 3 treatment periods for each patient. pVO_2 was defined as the average VO_2 during the last 5 seconds of peak exercise. The pVO_2 was also expressed as a percentage of predicted values (% predicted pVO_2), based on established reference equations.²² The respiratory exchange ratio (RER), defined as the rate of carbon dioxide produced divided by the oxygen used, was obtained at peak exercise and used to evaluate whether acceptable (RER >1.00) performance effort was achieved. The VO_2 at AT was determined with the use of the V-slope, and the VE/ VCO_2 slope was obtained at AT. The RPP (heart rate $\times$ systolic blood pressure) was calculated as a surrogate marker for myocardial work, that is, oxygen demand^{23–25} and O_2 -pulse (VO_2 /heart rate), as a surrogate marker of stroke volume.^{26,27}

Transthoracic echocardiography. A resting transthoracic echocardiography was performed by blinded investigators using Vivid E95 (General Electric) and analyzed with the use of EchoPAC software. To evaluate LVOT obstruction, we carefully analyzed the mitral valve movements for signs of systolic anterior motion, outflow tract turbulence, and mitral regurgitation. In addition, continuous-wave Doppler of the LVOT was performed to assess flow velocities at rest, during Valsalva maneuver, in a standing position, and standing after 1 minute of squatting. The gradients were automatically converted with the modified Bernoulli equation.

LV end-systolic and end-diastolic volumes were measured using the Simpson biplane method, and stroke volume and ejection fraction were derived from these values. LV GLS was assessed from 3 standard apical views by tracing the endocardium in end-diastole, using a 17-segment model. To assess LV diastolic function, pulsed-wave Doppler was placed at the tip of the mitral valve to obtain the transmitral flow velocities by assessing the early filling (E) and the atrial contraction (A), the E/A ratio, and the E deceleration time as surrogate markers of LV relaxation and stiffness.^{28–31} Tissue Doppler imaging, obtained from the mitral annulus at the septal, lateral, anterior, and inferior LV wall, was used for early diastolic tissue velocities (e'), and calculation of the E/ e' ratio. The e' velocities were used as markers of LV relaxation^{28,32} and the E/ e' ratio as a marker of LV filling pressures.^{28,31,33} Similarly, LAVI, assessed by biplane measurements, reflects the cumulative effects of increased LV filling pressures over time,^{28,34} and the TR pressure gradient, derived from continuous-wave Doppler with the Bernoulli equation, further correlates with left atrial pressures.^{28,30}

Symptoms, biomarkers, and cardiovascular examination. Symptom burden and health status were assessed using the patient-reported KCCQ Overall Summary Score (OSS) derived from the 23-item version. In addition, the NYHA functional class was evaluated by the investigator. The assessment also included biomarker testing of NT-proBNP and hs-cTnI, as well as 12-lead electrocardiography and automated blood pressure monitoring (average from 5 separate measurements over a 20-minute period).

STATISTICAL METHODS. Sample size calculation. In this triple cross-over design patients served as their own placebo control. To detect a minimal clinically relevant difference in the primary outcome, a pVO₂ difference (Δ pVO₂) of 1 mL/kg/min between bisoprolol and verapamil was assumed. Based on a paired *t*-test, an SD of 1.5 mL/kg/min, a 2-sided significance level of 5%, and 90% power, a sample size of 26 patients was required. The assumed SD of Δ pVO₂ was based on results from the TEMPO trial,²¹ where the SD of Δ pVO₂ was approximately 0.5 mL/kg/min. To account for potential differences between the TEMPO cohort and the present study population, the SD was multiplied by a factor of 3 in this sample size calculation. The sample size calculation was approved by Biostatistical Advisory Service at Aarhus University.

Treatment effects. Normality of data was assessed by means of QQ plots and histograms. Normally distributed data are presented as mean $\pm$ SD, non-normally distributed data as median (Q1-Q3), and categorical data as n (%). All analyses followed intention to treat; thus, all patients receiving the study drug, regardless of dosage, were included in the analysis.

To estimate the differences in treatment effect between bisoprolol, verapamil, and placebo on continuous and categorical variables, linear and ordinal mixed models were applied, respectively. We used mixed models with each outcome as the dependent variable, including fixed effects for treatment, period, and sequence, thus adjusting for the 2 latter, and a random effect for patient, thereby allowing each patient to serve as his or her own control. All 3 treatments were included in the model. For the prespecified primary outcome—pVO₂ between bisoprolol and verapamil—a prespecified 2-way analysis was performed. When evaluating all secondary outcomes, a factor-3 Bonferroni correction was applied to account for multiple testing between bisoprolol, verapamil, and placebo. Analyses were performed using all available observations, with the

mixed model accommodating missing outcome data. In case of nonnormally distributed residuals derived from the mixed model, a natural logarithmic transformation (ln) was applied, with subsequent conversion of the geometric mean value into the absolute scale using the relevant descriptive variable as a reference.

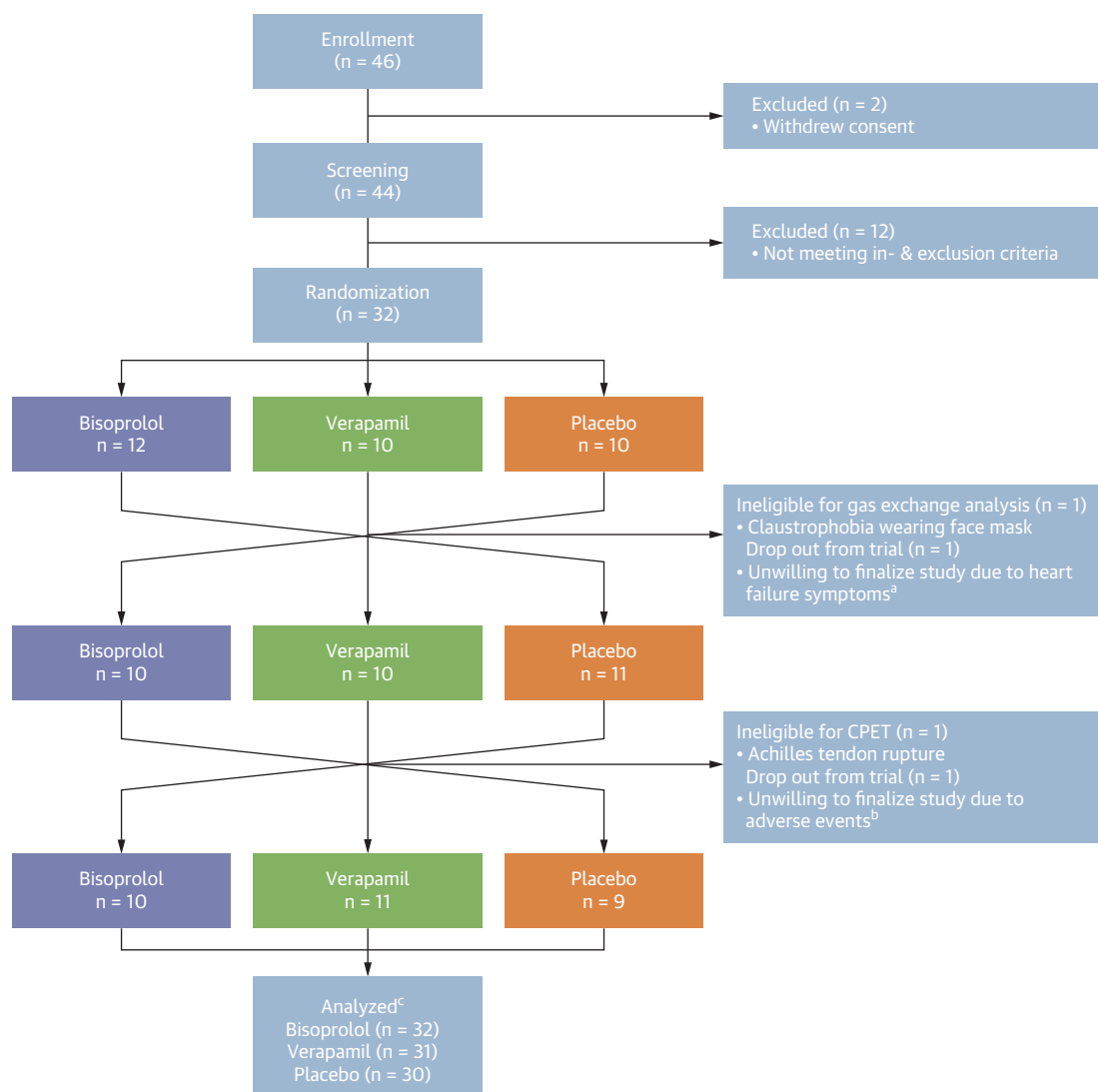
Correlations. To evaluate the correlation between heart rate and pVO₂ within each treatment, Pearson correlation coefficients (*r*) were calculated. Assumptions were assessed with the use of scatterplots, histograms, and QQ plots. Similar approaches were used when assessing the correlation between change in heart rate and change in pVO₂ between bisoprolol and placebo and between verapamil and placebo.

Carryover and period effects. To verify that neither carryover nor period effects influenced the outcomes (pVO₂, KCCQ-OSS, NYHA functional class, and NT-proBNP), we evaluated 3 components: 1) *carryover effect*, defined as the interaction between treatment (bisoprolol, verapamil, placebo) and sequence (order of administration) on the outcome; 2) *period effect*, defined as the interaction between treatment and period (treatment periods 1-3) on the outcome; and 3) *residual effect*, defined as any effect from the previous treatment influencing the outcome in the current treatment period. Carryover and period effects were tested by means of a Wald test (chi-square) within the mixed model. Residual effect was assessed in a separate mixed model including a “previous treatment” variable, tested in the same way with a Wald test.

A 2-sided 5% significance level was applied to all analyses. All statistical analyses were performed using Stata version 18.0.

RESULTS

BASELINE CHARACTERISTICS. From August 1, 2022, to September 3, 2024, 46 patients with non-obstructive HCM were enrolled by informed consent (Figure 2, Supplemental Results). Thirty-two patients fulfilled the inclusion and exclusion criteria and were randomly assigned to the study treatment. At the time of screening, 21 patients (66%) had NT-pro-BNP levels $\geq$ 300 ng/L, 15 (47%) were classified as NYHA functional class $\geq$ II, and 16 (50%) had a history of nonsustained ventricular tachycardia within 2 years before screening. A more detailed fulfillment of inclusion criteria is provided in Supplemental Table 3, which showed that 59% of patients met at least 2 of the 3 disease severity criteria. Patients had a mean age of 54 $\pm$ 15 years, and 11 (34%) were women (Table 1). Thirty-one (97%) patients had undergone

FIGURE 2 CONSORT Diagram

^aDid not receive verapamil in the second treatment period or placebo in the third treatment period. ^bDid not receive placebo in the third treatment period.

^c28 patients completed cardiopulmonary exercise testing (CPET) during all 3 treatment periods.

genetic testing for HCM on a clinical indication, of whom 52% carried a pathogenic or likely pathogenic sarcomere gene mutation. The KCCQ-OSS was 91.3 (Q1-Q3: 82.3-98.4) and 15 patients (47%) had NYHA functional class $\geq$ II at baseline. The maximum LV wall thickness was 20.0 ± 3.7 mm, LAVI was 64.3 ± 23.3 mL/m², and LV ejection fraction was $64 \pm 7\%$. The median NT-proBNP was 537 ng/L (Q1-Q3: 225-858 ng/L).

PRIMARY OUTCOME: pVO₂. Treatment with bisoprolol reduced the primary outcome of pVO₂ by -1.8 mL/kg/min (95% CI: -3.2 to -0.4 mL/kg/min;

$P = 0.013$) compared with verapamil (25.7 ± 8.7 mL/kg/min vs 28.2 ± 8.6 mL/kg/min, respectively) (Figure 3, Table 2).

Compared with placebo (28.7 ± 8.7 mL/kg/min), the reduction in pVO₂ was -2.5 mL/kg/min (95% CI: -4.2 to -0.7 mL/kg/min; $P = 0.002$) when treated with bisoprolol, whereas there was no significant difference between verapamil and placebo in pVO₂ (-0.7 mL/kg/min; 95% CI: -2.4 to 1.0 mL/kg/min; $P = 0.990$).

EXERCISE RESPONSE. Table 2 provides an overview of exercise response for bisoprolol, verapamil,

TABLE 1 Baseline Characteristics (N = 32)

Age, y	54.2 ± 14.9
Female	11 (34)
Body mass index, kg/m ²	26.7 (22.2-28.60)
HCM history	
Genitig testing for HCM (clinical indication)	31 (97)
Pathogenic or likely pathogenic gene mutation	16 (52)
MYH7	3 (19)
MYBPC3	4 (25)
TNNI3	3 (19)
TPM1	4 (25)
CSRP3	1 (6)
Family history of HCM	15 (47)
Family history of sudden cardiac death in family	6 (19)
History of NSVT <2 y before enrollment	16 (50)
HCM therapy	
Beta-blocker	10 (31)
Metoprolol	8 (80)
Bisoprolol	2 (20)
Verapamil	2 (6)
Implantable cardiac defibrillator	8 (25)
HCM symptoms	
NYHA functional class I	17 (53)
NYHA functional class II	13 (41)
NYHA functional class III	2 (6)
NYHA functional class IV	0 (0.0)
KCCQ-OSS	91.3 (82.3-98.4)
Palpitations	20 (63)
History of unexplained syncope	2 (6)
Comorbidities	
Hypertension	13 (41)
Diabetes	2 (6)
Paroxysmal atrial fibrillation	1 (3)
Chronic obstructive pulmonary disease	1 (3)
History of coronary revascularization	0 (0)
Vital signs	
Heart rate, beats/min	69±12
Systolic blood pressure, mm Hg	133±15
Diastolic blood pressure, mm Hg	82±10

Continued in the next column

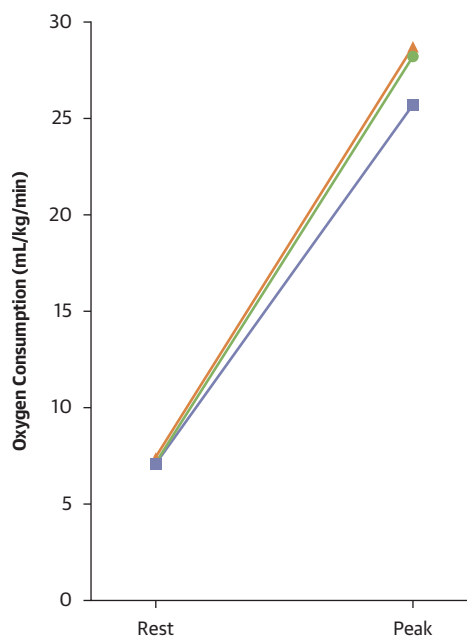
TABLE 1 Continued

Biomarkers	
hs-cTnI, ng/L	33 (17-58)
NT-proBNP, ng/L	537 (225-858)
Echocardiography	
IVSD, mm	17.4 ± 3.9
LVEDD, mm	45.7 ± 4.3
PWd, mm	10.8 ± 2.2
MWT, mm	20.0 ± 3.7
LVEF, %	64.4 ± 7.4
LAVI, mL/m ²	64.3 ± 23.3
Resting LVOT gradient, mm Hg	7.8 ± 3.1
Valsalva LVOT gradient, mm Hg	10.0 ± 5.4
Standing LVOT gradient, mm Hg	8.2 ± 4.7
Post-squat LVOT gradient, mm Hg	14.3 ± 6.0

Values are mean ± SD, n (%), or median (Q1-Q3).

CSRP3 = cysteine- and glycine-rich protein 3; HCM = hypertrophic cardiomyopathy; Hs-cTnI = high-sensitivity cardiac troponin I; IVSD = interventricular septum diameter KCCQ-OSS = Kansas City Cardiomyopathy Questionnaire Overall Summary Score; LAVI = left atrial volume index; LVEDD = left ventricular end-diastolic diameter; LVEF = left ventricular ejection fraction; LVOT = left ventricular outflow tract; MWT = maximal wall thickness; MYBPC3 = myosin-binding protein C3; MYH7 = myosin heavy chain 7; NSVT = nonsustained ventricular tachycardia; NT-proBNP = N-terminal pro-B-type natriuretic peptide; NPWd = posterior wall diameter; TNNI3 = troponin I3; TPM1 = tropomyosin alpha-1 chain.

FIGURE 3 Oxygen Consumption at Rest and at Peak Exercise



Bisoprolol (purple), verapamil (green), and placebo (orange).

and placebo treatments, and comparisons across treatments.

Bisoprolol and verapamil significantly reduced the heart rate at rest and peak exercise compared with placebo (Figure 4A). Notably, treatment with bisoprolol reduced the heart rate by −37 beats/min ($P < 0.001$) compared with placebo at peak exercise, whereas verapamil reduced the heart rate by −20 beats/min ($P < 0.001$).

At rest, both systolic and diastolic blood pressures were significantly reduced with bisoprolol treatment (−8 mm Hg [$P = 0.002$] and −7 mm Hg [$P < 0.001$],

TABLE 2 Treatment Effects of Bisoprolol, Verapamil, and Placebo on Cardiopulmonary Exercise Testing Parameters

	Bisoprolol (n = 32)	Verapamil (n = 31)	Placebo (n = 30)	Bisoprolol vs Verapamil		Bisoprolol vs Placebo		Verapamil vs Placebo		
				Adjusted Absolute Difference	P Value	Adjusted Absolute Difference	P Value	Adjusted Absolute Difference	P Value	
Resting CPET measurements										
Resting heart rate, beats/min	53 ± 9	66 ± 12	73 ± 12	−13 (−17 to −9)	<0.001	−19 (−23 to −15)	<0.001	−6 (−10 to −2)	0.003	
Resting systolic BP, mm Hg	122 ± 15	126 ± 14	132 ± 15	−3 (−9 to 3)	0.532	−8 (−14 to −3)	0.002	−5 (−11 to 1)	0.099	
Resting diastolic BP, mm Hg	74 ± 10	76 ± 10	82 ± 11	−2 (−6 to 2)	0.516	−7 (−11 to −3)	<0.001	−5 (−9 to −1)	0.010	
Resting RER	0.78 ± 0.06	0.79 ± 0.07	0.80 ± 0.09	−0.01 (−0.04 to 0.03)	1.000	−0.02 (−0.05 to 0.02)	0.814	−0.01 (−0.04 to 0.02)	1.000	
Resting VO ₂ , mL/kg/min	7.1 ± 2.2	7.1 ± 1.7	7.4 ± 1.9	−0.08 (−1.20 to 1.03)	1.000	−0.41 (−1.52 to 0.70)	1.000	−0.33 (−1.44 to 0.78)	1.000	
Resting RPP, beats/min•mm Hg	6,557 ± 1,313	8,349 ± 1,681	9,585 ± 2,028	−1,793 (−2,469 to −1,116)	<0.001	−3,005 (−3,89 to −2,320)	<0.001	−1,212 (−1,899 to −525)	<0.001	
Resting O ₂ -pulse, mL/beat	11.40 ± 3.91	9.20 ± 3.03	8.74 ± 2.63	2.17 (0.60-3.73)	0.003	2.73 (1.16 to 4.30)	<0.001	0.56 (−1.00 to 2.12)	1.000	
Submaximal CPET measurements										
VO ₂ at AT, mL/kg/min	13.59 ± 4.27	13.58 ± 4.81	14.89 ± 4.87	−0.04 (−1.63 to 1.56)	1.000	−1.26 (−2.87 to 0.34)	0.181	−1.23 (−2.80 to 0.35)	0.186	
VE/VCO ₂ slope	32.34 ± 5.73	32.06 ± 5.49	31.51 ± 5.23	0.38 (−1.14 to 1.90)	1.000	0.73 (−0.81 to 2.26)	0.769	0.35 (−1.17 to 1.87)	1.000	
Peak CPET measurements										
Exercise time, min:s	09:47 ± 03:03	10:17 ± 03:10	10:41 ± 03:08	−00:25 (−00:53 to 00:03)	0.089	−00:47 (−01:14 to −00:18)	<0.001	−00:21 (−00:44 to 00:07)	0.202	
Peak workload, W	163 ± 59	173 ± 64	179 ± 65	−6.5 (−13.2 to −0.2)	0.058	−11.5 (−18.3 to −4.8)	<0.001	−5.0 (−11.7 to 1.7)	0.216	
Peak RER	1.10 ± 0.10	1.08 ± 0.09	1.10 ± 0.10	0.01 (−0.02 to 0.03)	1.000	−0.01 (−0.03 to 0.02)	1.000	−0.02 (−0.04 to 0.01)	0.189	
Peak heart rate, beats/min	120 ± 23	141 ± 19	157 ± 19	−20 (−27 to −13)	<0.001	−37 (−44 to −30)	<0.001	−17 (−24 to −11)	<0.001	
Peak systolic BP, mm Hg	172 ± 42	175 ± 30	180 ± 39	−4 (−24 to 16)	1.000	−9 (−29 to 11)	0.831	−5 (−26 to 15)	1.000	
Peak diastolic BP, mm Hg	75 ± 22	87 ± 23	90 ± 24	−13 (−26 to 0)	0.045	−16 (−29 to −3)	0.007	−3 (−16 to 10)	1.000	
Peak RPP, ^b beats/min•mm Hg	20,895 ± 7,139	24,644 ± 4,874	27,955 ± 6,182	−3,744 (−7,231 to −257)	0.030	−7,234 (−10,729 to −3,739)	<0.001	−3,490 (−6,998 to 18)	0.052	
Peak VO ₂ , mL/kg/min	25.7 ± 8.7	28.2 ± 8.6	28.7 ± 8.7	−1.8 (−3.2 to −0.4) ^a	0.013 ^a	−2.5 (−4.2 to −0.7)	0.002	−0.7 (−2.4 to 1.0)	0.990	
Peak % predicted VO ₂ ^b	82 ± 30	90 ± 31	93 ± 30	−8 (−14 to −2)	0.004	−8.9 (−14.8 to −3.1)	0.001	−1.2 (−7.1 to 4.6)	1.000	
Peak O ₂ -pulse, mL/beat	18.46 ± 5.13	16.98 ± 4.48	15.57 ± 3.90	1.75 (0.65-2.85)	<0.001	3.30 (2.21-4.40)	<0.001	1.55 (0.49-2.62)	0.001	

Ranges in parentheses are 95% CIs. Treatment effects were estimated with the use of a linear mixed model, adjusting for period, sequence, and individual effects. Bonferroni correction was applied for multiple comparisons. The adjusted mean differences derive from these models. ^aDerived from the 2-way analysis as described in the Methods. ^bManually calculated with the use of reference values of maximal oxygen consumption at different ages from Fletcher et al.²²

AT = anaerobic threshold; BP = blood pressure; CPET = cardiopulmonary exercise testing; O₂-pulse = VO₂/heart rate as an estimate for stroke volume; RER = respiratory exchange ratio; RPP = rate-pressure product = heart rate × systolic blood pressure as an estimate for myocardial workload; VE/VCO₂ = minute ventilation/CO₂ production as an estimate for ventilatory efficiency; VO₂ = oxygen consumption.

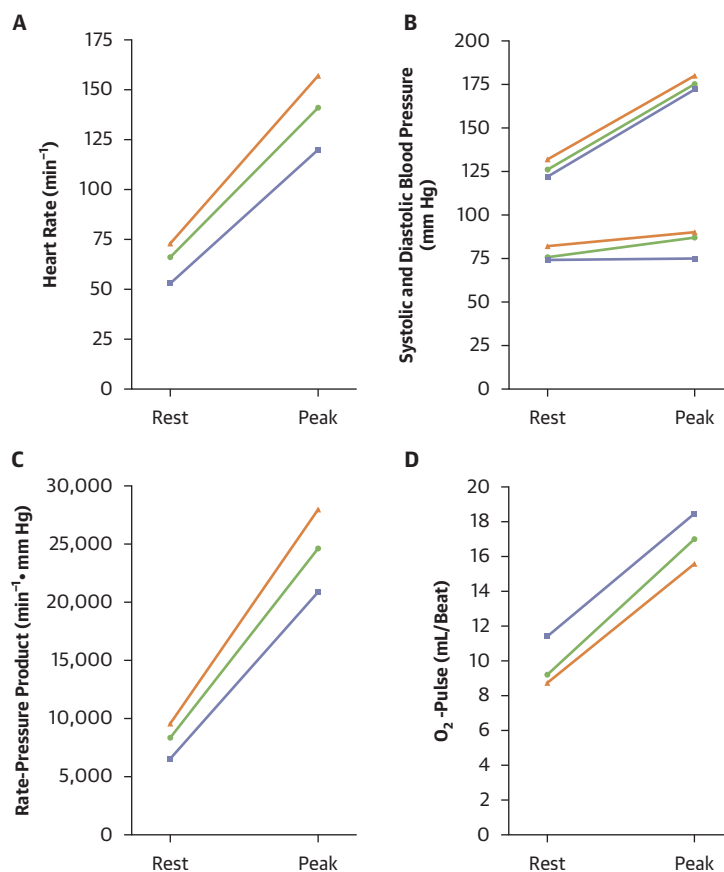
respectively), although verapamil affected only diastolic blood pressure (–5 mm Hg; $P = 0.010$) significantly compared with placebo (Figure 4B). At peak exercise, only diastolic blood pressure was reduced when treated with bisoprolol (–16 mm Hg, $P = 0.007$) compared with placebo.

RPP, as a marker of myocardial work, also was reduced with bisoprolol and verapamil at rest

(–3,005 beats/min·mm Hg [$P < 0.001$] and –1,212 beats/min·mm Hg [$P < 0.001$], respectively) and at peak exercise (–7,234 beats/min·mm Hg [$P < 0.001$] and –3,491 beats/min·mm Hg [$P = 0.052$]) compared with placebo, indicating a potential decrease in oxygen demand (Figure 4C).

O₂-pulse, as a marker of stroke volume, was significantly increased with bisoprolol both at rest

FIGURE 4 Exercise Parameters at Rest and Peak Exercise



(A) Heart rate, (B) systolic and diastolic blood pressure, (C) rate-pressure product, and (D) O₂-pulse. Bisoprolol (purple), verapamil (green), and placebo (orange).

(+2.73 mL/beat; $P < 0.001$) and at peak exercise (+3.30 mL/beat; $P < 0.001$) compared with placebo, although treatment with verapamil increased O₂-pulse significantly only at peak exercise (+1.55 mL/beat; $P = 0.001$) (Figure 4D).

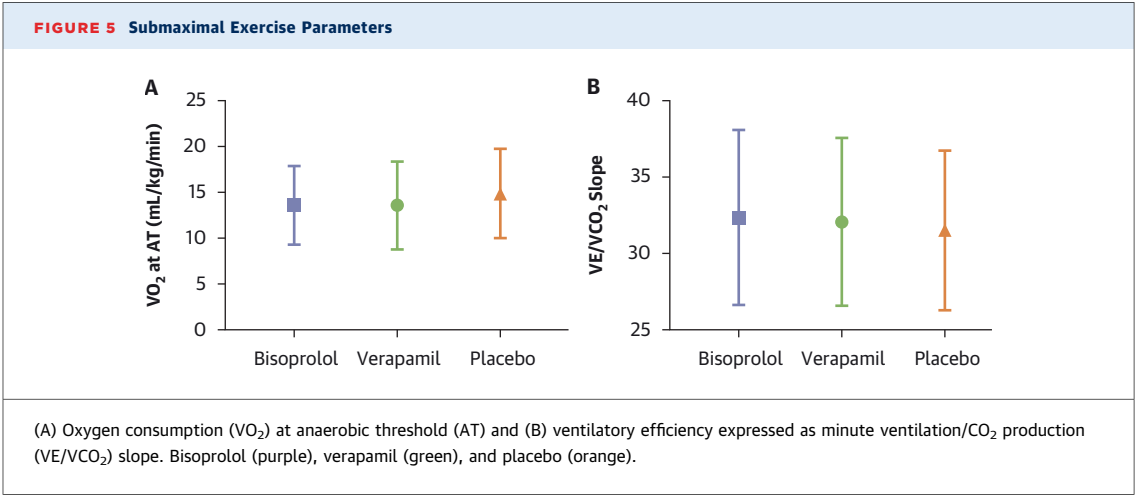
Neither bisoprolol or verapamil changed VO₂ at AT (−1.26 mL/kg/min [$P = 0.181$] and 1.23 mL/kg/min [$P = 0.186$], respectively) (Figure 5A) or VE/VCO₂ slope (+0.73 [$P = 0.769$] and +0.35 [$P = 1.000$], respectively) (Figure 5B) compared with placebo.

CORRELATION BETWEEN HEART RATE RESPONSE AND PEAK VO₂. There was a significant, yet moderate, correlation between heart rate and VO₂ during peak exercise with bisoprolol treatment ($r = 0.496$; $P = 0.006$) and verapamil ($r = 0.446$; $P = 0.013$). In contrast, there was no significant correlation during placebo treatment ($r = 0.258$; $P = 0.169$) (Figure 6A). When analyzing the correlation between change in

peak heart rate and change in pVO₂ between bisoprolol and placebo ($r = 0.101$; $P = 0.370$) (Figure 6B) and between verapamil and placebo ($r = 0.153$; $P = 0.157$) (Figure 6C), there was no significant correlations.

LV VOLUMES AND SYSTOLIC FUNCTION. Table 3 provides an overview of echocardiographic measures for bisoprolol, verapamil, and placebo treatments, and comparisons across treatment.

Treatment with bisoprolol increased the LV end-diastolic and end-systolic volumes (+13.4 mL [$P = 0.004$] and +8.1 mL [$P = 0.001$], respectively), whereas verapamil did not affect these parameters (+5.8 mL [$P = 0.482$] and +3.2 mL [$P = 0.467$], respectively) compared with placebo. Despite these changes in LV volumes, the stroke volume and ejection fraction did not change significantly with bisoprolol (+5.2 mL [$P = 0.198$] and −1.8% [$P = 0.496$],



respectively) or verapamil (+2.6 mL [$P = 1.000$] and -1.4% [$P = 0.848$], respectively). Treatment with verapamil improved LV GLS (-1.1% ; $P = 0.001$), although bisoprolol did not demonstrate any significant changes (-0.6% ; $P = 0.263$) compared with placebo. However, there were no differences in LV ejection fraction between treatments.

LV DIASTOLIC FUNCTION. Compared with placebo, treatment with bisoprolol increased the LAVI by $+13.0 \text{ mL/m}^2$ ($P < 0.001$), whereas verapamil demonstrated no significant changes ($+5.2 \text{ mL/m}^2$; $P = 0.194$) (Figure 7A). Both bisoprolol and verapamil increased peak E-wave velocity ($+6.5 \text{ cm/s}$ [$P = 0.006$] and $+9.2 \text{ cm/s}$ [$P < 0.001$], respectively) and reduced peak A-wave velocity (-17.4 cm/s

[$P < 0.001$] and -7.5 cm/s [$P = 0.026$], respectively) resulting in a significantly higher E/A ratio ($+0.42$ [$P < 0.001$] and $+0.27$ [$P < 0.001$], respectively) compared with placebo (Table 3). None of the treatments changed the E deceleration time. Treatment with bisoprolol did not affect the mitral annulus diastolic velocities (e') at the septal ($+0.34 \text{ cm/s}$; $P = 0.460$) or lateral (0.19 cm/s ; $P = 1.000$) walls or septal or lateral E/ e' ratios ($+0.35$ [$P = 1.000$] and 1.14 [$P = 0.117$], respectively) (Figures 7B and 7C). Verapamil increased e' velocity by $+1.39 \text{ cm/s}$ ($P < 0.001$) in the lateral wall and $+1.19 \text{ cm/s}$ ($P < 0.001$) in the septal wall compared with placebo. However, there were no differences in lateral and septal E/ e' ratios (-0.60 [$P = 0.487$] and -0.18 [$P = 1.000$],

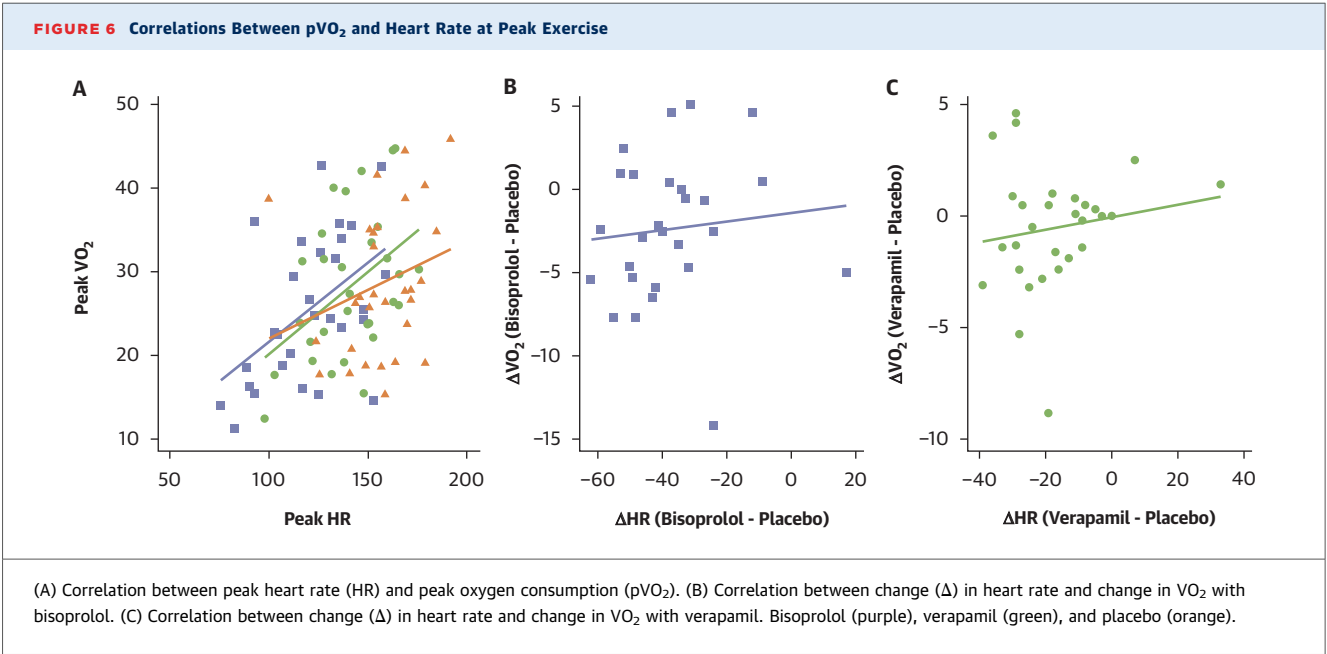


TABLE 3 Treatment Effects of Bisoprolol, Verapamil, and Placebo on Echocardiographic Parameters

	Bisoprolol (n = 32)	Verapamil (n = 31)	Placebo (n = 30)	Bisoprolol vs Verapamil		Bisoprolol vs Placebo		Verapamil vs Placebo		
				Adjusted Absolute Difference (95% CI)	P Value	Adjusted Absolute Difference (95% CI)	P Value	Adjusted Absolute Difference (95% CI)	P Value	
LV volumes and systolic function										
LV stroke volume, mL	65.2 ± 16.2	62.8 ± 20.1	60.7 ± 16.3	2.6 (−4.1 to 9.2)	1.000	5.2 (−1.6 to 11.9)	0.198	2.6 (−4.2 to 9.4)	1.000	
LV ejection fraction, %	62.4 ± 7.7	62.7 ± 6.8	64.3 ± 7.2	−0.4 (−3.5 to 2.7)	1.000	−1.8 (−4.9 to 1.3)	0.496	−1.4 (−4.5 to 1.7)	0.848	
LV GLS, %	−15.3 ± 3.2	−15.7 ± 3.5	−14.7 ± 3.8	0.4 (−0.4 to 1.3)	0.653	−0.6 (−1.5 to 0.3)	0.263	−1.1 (−1.9 to −0.2)	0.011	
LV end-diastolic volume, mL	107.1 ± 31.6	100.2 ± 30.4	94.9 ± 24.1	7.7 (−2.2 to 17.36)	0.199	13.4 (3.4 to 23.3)	0.004	5.8 (−4.1 to 15.8)	0.482	
LV end-systolic volume, mL	41.9 ± 18.6	37.4 ± 13.3	34.2 ± 11.6	4.9 (−0.3 to 10.2)	0.074	8.1 (2.8 to 13.4)	0.001	3.2 (−2.17 to 8.5)	0.467	
LV diastolic function										
LA volume index, mL/m ²	74.9 ± 26.5	67.6 ± 23.7	63.0 ± 21.9	7.8 (1.2-14.5)	0.014	13.0 (6.3-19.8)	<0.001	5.2 (−1.5 to 11.9)	0.194	
Peak E-wave velocity, cm/s	64.5 ± 20.5	67.1 ± 20.0	58.8 ± 19.4	−2.8 (−7.8 to 2.2)	0.549	6.5 (1.4-11.5)	0.006	9.2 (4.2-14.3)	<0.001	
Peak A-wave velocity, cm/s	51.5 ± 17.6	60.1 ± 22.0	68.0 ± 24.0	−9.8 (−16.6 to 3.0)	0.002	−17.4 (−24.2 to −10.5)	<0.001	−7.5 (−14.4 to −0.7)	0.026	
Peak E/A	1.24 (0.97-1.45)	1.07 (0.83-1.36)	0.89 (0.71-1.09)	0.10 (−0.03 to 0.21)	0.178	0.42 (0.23-0.63) ^a	<0.001 ^a	0.27 (0.11-0.46) ^a	<0.001 ^a	
E deceleration time, ms	208.3 ± 64.4	210.2 ± 56.7	222.2 ± 67.4	−2.7 (−29.9 to 24.4)	1.000	−15.5 (−43.0 to 12.0)	0.530	−12.8 (−40.4 to 14.8)	0.802	
Anterior e′ velocity, cm/s	7.03 ± 2.93	8.03 ± 3.67	6.72 ± 3.42	−0.88 (−1.71 to −0.14)	0.013	0.63 (−0.10 to 1.44) ^a	0.123 ^a	1.59 (0.77-2.51) ^a	<0.001 ^a	
Inferior e′ velocity, cm/s	6.27 ± 1.87	7.07 ± 2.21	8.41 ± 10.23	−0.11 (−2.90 to 0.54)	0.386	−0.77 (−2.05 to 0.76) ^a	0.627 ^a	−0.16 (−1.28 to 1.88) ^a	1.000 ^a	
Lateral e′ velocity, cm/s	7.91 ± 3.30	9.23 ± 3.77	7.93 ± 3.81	−1.19 (−2.05 to −0.34)	0.002	−0.19 (−0.67 to 1.06)	1.000	1.39 (0.51 to 2.25)	<0.001	
Septal e′ velocity, cm/s	6.19 ± 2.01	7.13 ± 2.46	6.03 ± 1.97	−0.85 (−1.41 to −0.28)	0.001	0.34 (−0.23 to 0.91)	0.460	1.19 (0.61 to 1.76)	<0.001	
Lateral E/e′	9.85 ± 5.51	8.10 ± 3.43	9.12 ± 4.74	0.92 (−0.08 to 1.81)	0.082	0.35 (−0.69 to 1.52) ^a	1.000 ^a	−0.60 (−1.54 to 0.45) ^a	0.487 ^a	
Septal E/e′	11.48 ± 4.77	9.9 ± 3.86	10.2 ± 3.3	1.32 (0.01 to 2.62)	0.047	1.14 (−0.18 to 2.46)	0.117	−0.18 (−1.50 to 1.15)	1.000	
Right ventricular function										
TAPSE, mm	24.4 ± 4.1	25.6 ± 4.4	24.0 ± 3.6	−1.1 (−2.8 to 0.5)	0.311	0.6 (−1.1 to 2.2)	1.000	1.7 (0.0-3.4)	0.048	
TR pressure gradient, mm Hg	20.5 ± 9.9	17.2 ± 11.7	16.1 ± 8.9	3.2 (−1.0 to 7.5)	0.211	4.3 (0.0-8.6)	0.049	1.1 (−3.3 to 5.4)	1.000	
Descriptive values are presented as mean ± SD or median (Q1-Q3). Treatment effects were estimated with the use of a linear mixed model adjusting for period, sequence, and individual effects. ^a Derived from natural log-transformation (ln) and conversion of the geometric mean ratio to an absolute scale using the mean/median of placebo as reference from descriptive data. A = atrial mitral inflow velocity; E = early mitral inflow velocity; e′ = early diastolic mitral annular velocity; GLS = global longitudinal strain; LA = left atrial; LV = left ventricular; NT-proBNP = N-terminal pro-B-type natriuretic peptide; TAPSE = tricuspid annular plane systolic excursion; TR = tricuspid regurgitation.										

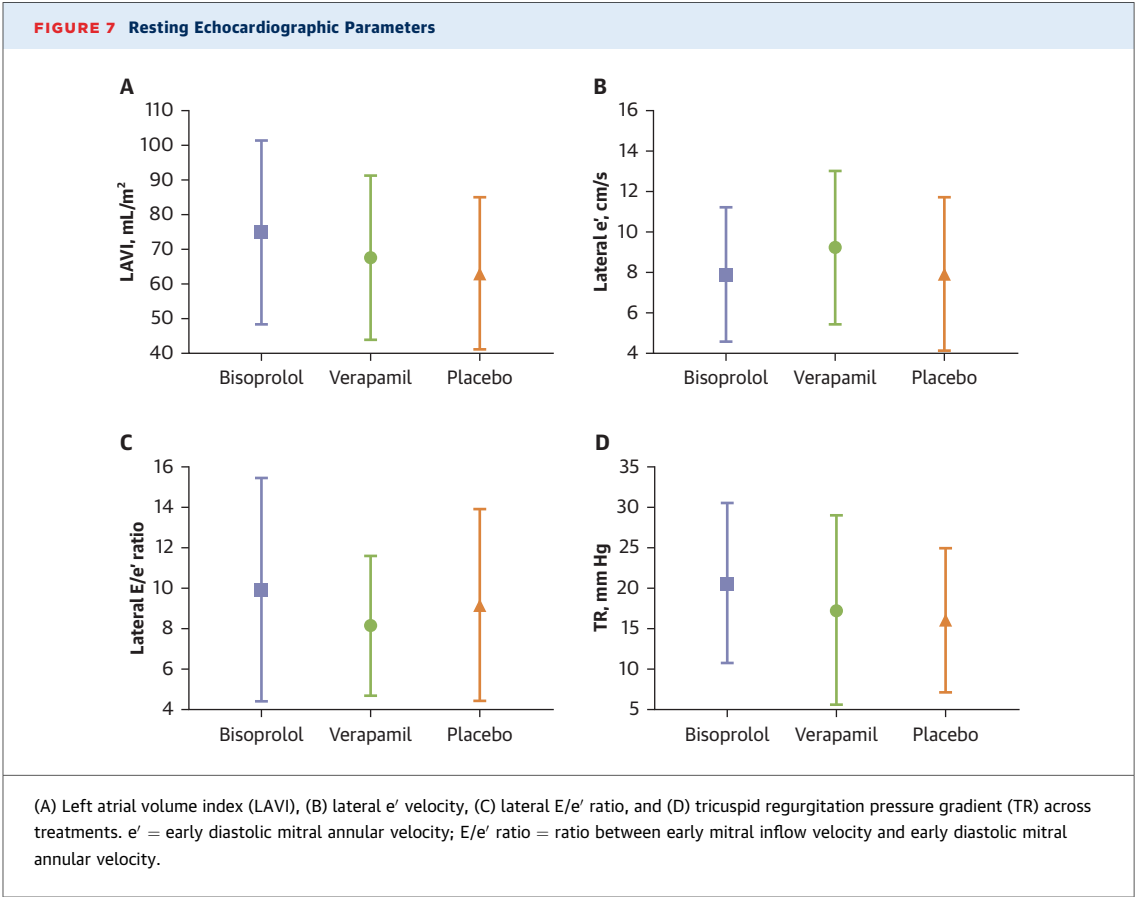
respectively) between verapamil and placebo. Furthermore, the TR pressure gradient increased by +4.3 mm Hg ($P = 0.049$) with bisoprolol treatment, although verapamil showed no change (+1.1 mm Hg; $P = 1.000$) compared with placebo (Figure 7D).

Table 4 provides an overview of the directional statistically significant effects of bisoprolol and verapamil on the primary diastolic parameters compared with placebo. In general, bisoprolol treatment either worsened the diastolic parameters or left them unchanged compared with placebo, whereas the signal with verapamil was more mixed. When categorizing diastolic parameters as “improved,”

“worsened,” and “unchanged” for each patient, the overall diastolic profile of the treatments seems to be in favor of verapamil, with a larger proportion of patients improving their diastolic parameters with verapamil (Figure 8).

SYMPTOMS. Table 5 provides an overview of symptoms and biomarkers for bisoprolol, verapamil, and placebo treatment, and comparisons across treatments.

When treated with bisoprolol, patients reported a worsening of health status, as reflected by the reduced KCCQ-OSS of −6.6 points ($P = 0.001$) compared with placebo (Figure 9A). Treatment with verapamil did not change KCCQ-OSS (−2.3 points;



$P = 0.668$) significantly compared with placebo. There were no differences in NYHA functional classification between any of the treatments (Figure 9B).

BIOMARKERS. Treatment with bisoprolol increased NT-proBNP by 165 ng/L ($P = 0.006$), percentage change of +50.8% (Q1-Q3: -1.6% to 79.8%), and

lowered hs-cTnI by -4.6 ng/L ($P < 0.001$), compared with placebo (Table 5, Figure 10). In contrast, verapamil treatment reduced NT-proBNP by -177 ng/L ($P < 0.001$), percentage change of -35.6% (Q1-Q3: -51.6% to -2.0%), and hs-cTnI by -5.4 ng/L ($P < 0.001$) compared with placebo.

ADVERSE EVENTS. One serious adverse event (Achilles tendon rupture) unrelated to the study drug was observed. Adverse events were reported in 19 patients (59%) during bisoprolol, 17 patients (55%) during verapamil, and 14 patients (47%) during placebo treatment (Supplemental Table 4). Dose reduction ($n = 6$) during bisoprolol treatment was due to exhaustion ($n = 5$), dizziness ($n = 3$), headaches ($n = 2$), palpitations ($n = 2$), and single reports of gastrointestinal discomfort, dyspnea or cold extremities. During verapamil treatment, dose reduction ($n = 2$) was due to combined gingival hyperplasia, dyspnea, and headaches in one patient, and another reported exhaustion, bradycardia, and hypotension. During placebo treatment, dose reduction ($n = 2$) was due to combined dizziness, palpitation, and gastrointestinal discomfort in one patient, and another reported exhaustion.

TABLE 4 Overview of Changes in Diastolic Parameters During Bisoprolol and Verapamil Treatments Compared With Placebo		
	Bisoprolol	Verapamil
LAVI	↑	→
Peak E-wave velocity	↑	↑
Lateral e'-velocity	→	↑
Septal e' velocity	→	↑
Lateral E/e' ratio	→	→
Septal E/e' ratio	→	→
TR pressure gradient	↑	→
NT-proBNP	↑	↓

Upward arrows (↑) represent a significant increase and downward arrows (↓) a significant reduction; horizontal arrows (→) indicate no significant difference compared with placebo treatment.

LAVI = left atrial volume index; E = early mitral inflow velocity; e' = early diastolic mitral annular velocity; TR = tricuspid regurgitation; NT-proBNP = N-terminal pro-B-type natriuretic peptide.

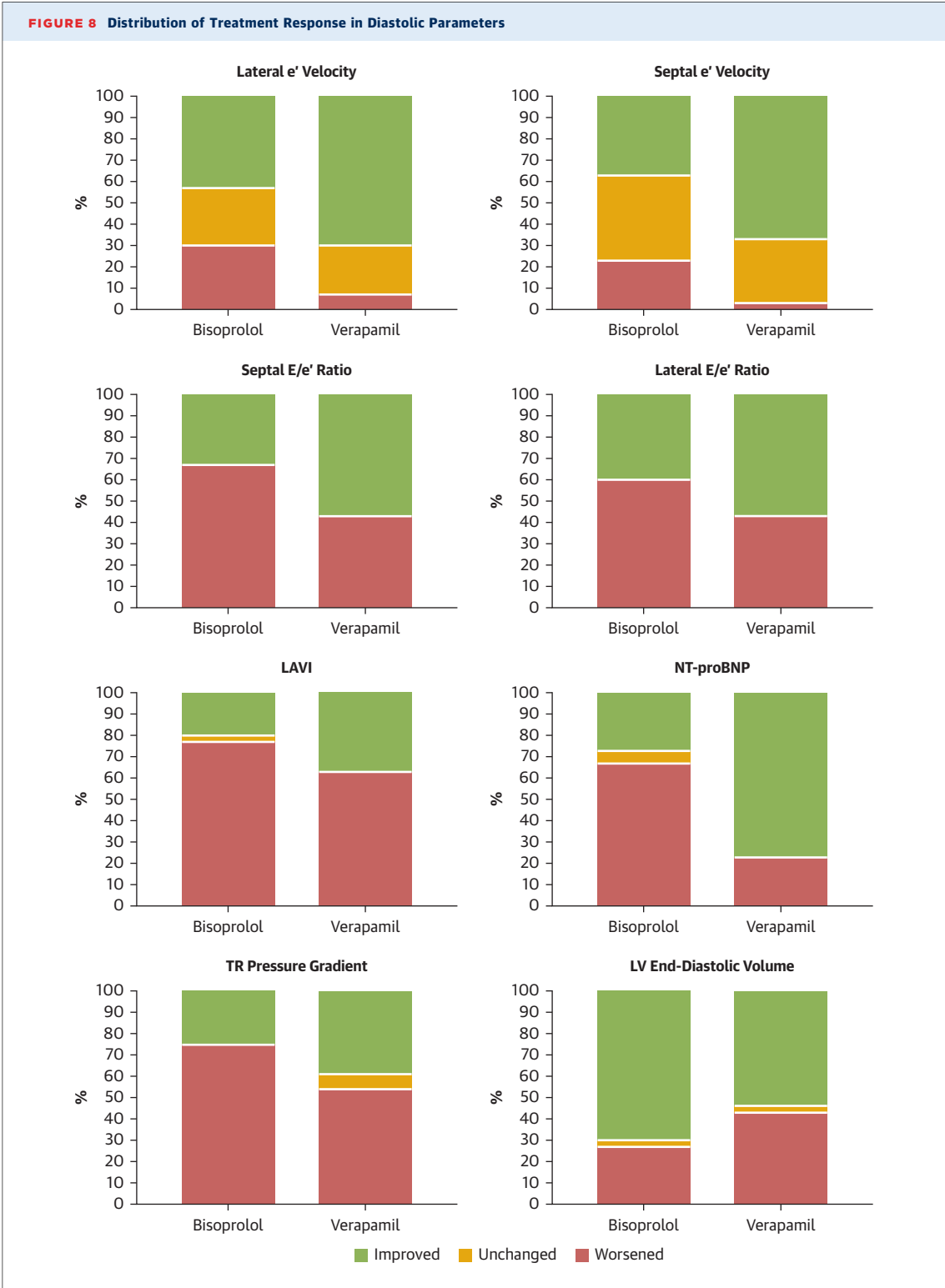


Illustration of the number of patients experiencing an improvement (green), worsening (red), or remaining unchanged (orange) with bisoprolol and verapamil treatment compared with placebo. e' = early diastolic mitral annular velocity; E/e' ratio = ratio between early mitral inflow velocity and early diastolic mitral annular velocity; LAVI = left atrial volume index; LV = left ventricular; NT-proBNP = N-terminal pro-B-type natriuretic peptide; TR = tricuspid regurgitation.

TABLE 5 Treatment Effects of Bisoprolol, Verapamil, and Placebo on Symptoms and Biomarkers

	Bisoprolol (n = 32)	Verapamil (n = 31)	Placebo (n = 30)	Bisoprolol vs Verapamil		Bisoprolol vs Placebo		Verapamil vs Placebo	
				Adjusted Absolute Difference (95% CI)	P Value	Adjusted Absolute Difference (95% CI)	P Value	Adjusted Absolute Difference (95% CI)	P Value
KCCQ-OSS	90 (77-98)	93 (83-98)	94 (90-98)	-4.3 (-8.8 to 0.1)	0.056	-6.6 (-11.1 to -2.2)	0.001	-2.3 (-6.8 to 2.2)	0.668
NYHA functional class	-	-	-	-	0.940	-	1.000	-	1.000
II	20 (63)	16 (52)	18 (60)	-	-	-	-	-	-
III	9 (28)	13 (41)	8 (27)	-	-	-	-	-	-
NT-proBNP, ng/L	578 (318-1,737)	330 (165-861)	554 (243-1,016)	263 (199-315) ^a	<0.001 ^a	165 (33-326) ^a	0.006 ^a	-177 (-245 to -92) ^a	<0.001 ^a
hs-cTnI, ng/L	28 (13-52)	26 (11-53)	27 (15-49)	1.0 (-2.1 to 3.8) ^a	1.000 ^a	-4.6 (-7.0 to -1.9) ^a	<0.001 ^a	-5.4 (-7.7 to -2.8) ^a	<0.001 ^a

Descriptive values are presented as mean ± SD or median (Q1-Q3). Treatment effects were estimated with the use of a linear or ordinal mixed model, adjusting for period, sequence, and individual effects. Bonferroni correction was applied for multiple comparisons. The adjusted mean differences derive from these models. ^aDerived from natural log-transformation (ln) and conversion of the geometric mean ratio to an absolute scale using the median of placebo as reference from descriptive data.

Hs-cTnI = high-sensitivity cardiac troponin I; KCCQ-OSS = Kansas City Cardiomyopathy Questionnaire Overall Summary Score; NT-proBNP = N-terminal pro-B-type natriuretic peptide.

Two patients dropped out of the study. One withdrew after experiencing gingival hyperplasia, dyspnea, and headaches during verapamil treatment in the first period, followed by dizziness, dyspnea, and exhaustion during bisoprolol treatment in the second period. Another patient, whose regular BB treatment was discontinued upon trial enrollment, withdrew consent owing to palpitations and dyspnea (Figure 2).

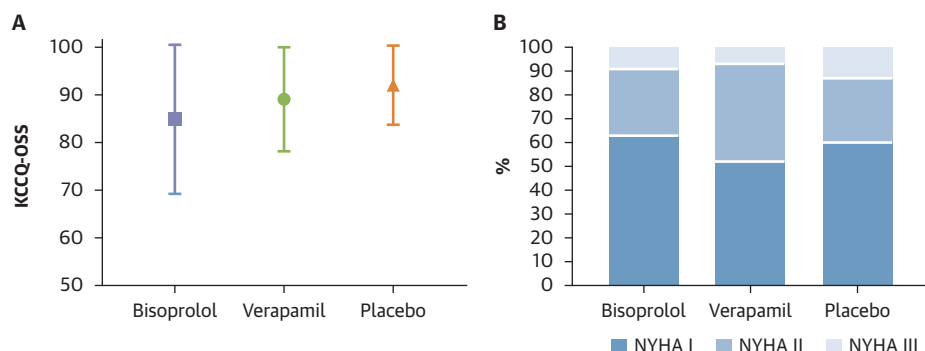
STUDY DRUG TOLERABILITY. Distribution of doses across the 3 treatments is presented in Figure 11. Out of the 32 patients completing the bisoprolol treatment period, 26 patients received the target dose of 7.5 mg bisoprolol per day, 3 patients received 5 mg, 1 received 2.5 mg, and 2 patients did not tolerate bisoprolol owing to side-effects and worsening of heart failure symptoms and were titrated to 0 mg.

Out of 31 patients completing the verapamil treatment period, 29 patients received the target dose of 360 mg verapamil per day, 1 patient received 240 mg, and 1 patient did not tolerate verapamil owing to side-effects and was titrated to 0 mg.

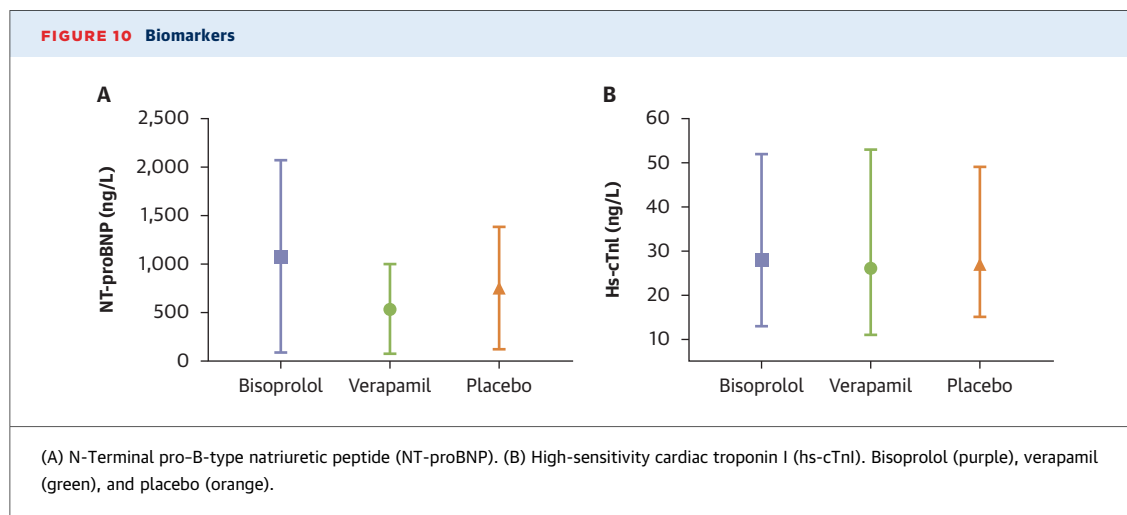
Out of 30 patients completing the placebo treatment period, 28 patients received the target dose of 3 placebo capsules per day, 2 received 2 capsules, and no patients received 1 or 0 capsules.

CARRYOVER AND PERIOD EFFECT. There was no evidence of carryover or period effects when evaluating the effect of treatment on pVO₂, KCCQ-OSS, NYHA functional class, or NT-proBNP (all *P* > 0.634). Likewise, the additional analyses revealed no residual effects (all *P* > 0.29) (Table 6).

FIGURE 9 KCCQ-OSS and NYHA Functional Classification Across Treatments



(A) Kansas City Cardiomyopathy Questionnaire Overall Summary Score (KCCQ-OSS). Bisoprolol (purple), verapamil (green), and placebo (orange). (B) Distribution of NYHA functional classification across treatments.



DISCUSSION

This is the first randomized, double-blind, placebo-controlled trial to compare the effects of bisoprolol and verapamil, head-to-head, in a well defined exclusively nonobstructive HCM cohort. The study provides novel insight into the effects of conventional treatments on functional capacity, exercise response, myocardial function, structural changes, symptoms, and biomarkers (**Central Illustration**).

EXERCISE RESPONSE. We demonstrated that treatment with bisoprolol was associated with reduced functional capacity as assessed by pVO_2 compared

with both verapamil and placebo. In comparison, verapamil did not change functional capacity compared with placebo.

As a direct correlate of cardiac output, pVO_2 is a well established independent predictor of clinical outcomes in HCM.³⁵ The reduction in pVO_2 with bisoprolol exceeded thresholds previously associated with clinical outcomes.³⁶ Our findings align with treatment effects on exercise capacity reported in 1993 by Gilligan et al,¹⁸ who assessed the effect of nadolol, a nonselective beta-receptor antagonist, verapamil, and placebo in a triple-crossover study. However, the trial was inconclusive, likely owing to lack of statistical power and inclusion of a cohort of 18 patients with a mix of obstructive and non-obstructive HCM.¹⁸

A potential mechanism for the observed differences in pVO_2 between treatments could relate to differences in heart rate response. We showed that treatment with bisoprolol was associated with a greater reduction in heart rate (−37 beats/min) than verapamil (−17 beats/min) during peak exercise. Subsequently, correlation analysis demonstrated a moderate correlation between peak heart rate and pVO_2 with both bisoprolol and verapamil, illustrating that patients with low peak heart rate tended to have lower pVO_2 . This correlation was not significant for placebo, suggesting that in patients with non-obstructive HCM, without treatment with negative chronotropic drugs, other factors than heart rate alone contribute to the limitation of pVO_2 .

Patients with nonobstructive HCM share several characteristics with patients with heart failure with preserved ejection fraction (HFpEF). Previous studies have demonstrated negative impact on chronotropic competence and functional capacity in HFpEF,

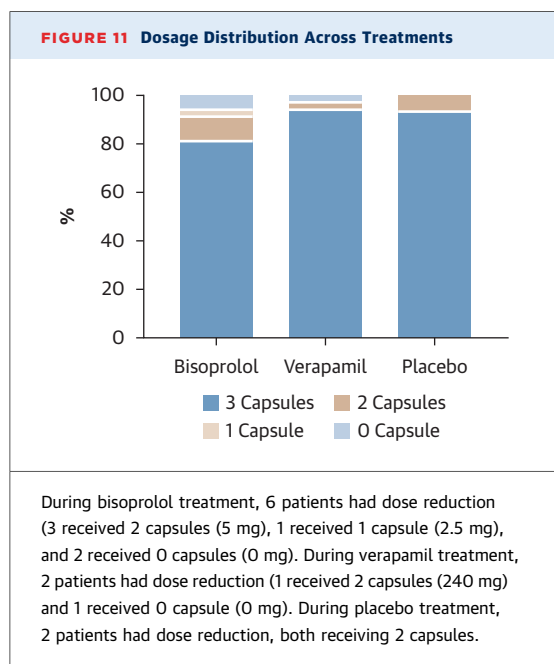


TABLE 6 Carryover and Period Effect

	Carryover Effects		Period Effects		Residual Effects	
	Chi-Square	P Value	Chi-Square	P Value	Chi-Square	P Value
pVO ₂	3.55	0.894	1.12	0.890	1.24	0.539
KCCQ-OSS	1.75	0.988	1.03	0.905	2.24	0.326
NYHA	3.70	0.883	2.37	0.669	NA	NA
NT-proBNP	6.12	0.634	2.33	0.675	2.47	0.291

The carryover effect assesses the interaction between treatment (bisoprolol, verapamil, placebo) and sequence (order of administration) on the outcome. The period effect assesses the interaction between treatment and the period in which it was administered (treatment periods 1–3) on the outcome. Residual effect evaluates whether the treatment from the preceding treatment period influenced the outcome in the current treatment period. All effects were tested using a Wald test (χ^2) within the mixed model.

KCCQ-OSS = Kansas City Cardiomyopathy Questionnaire Overall Summary Score; NT-proBNP = N-terminal pro-B-type natriuretic peptide; pVO₂ = peak oxygen consumption.

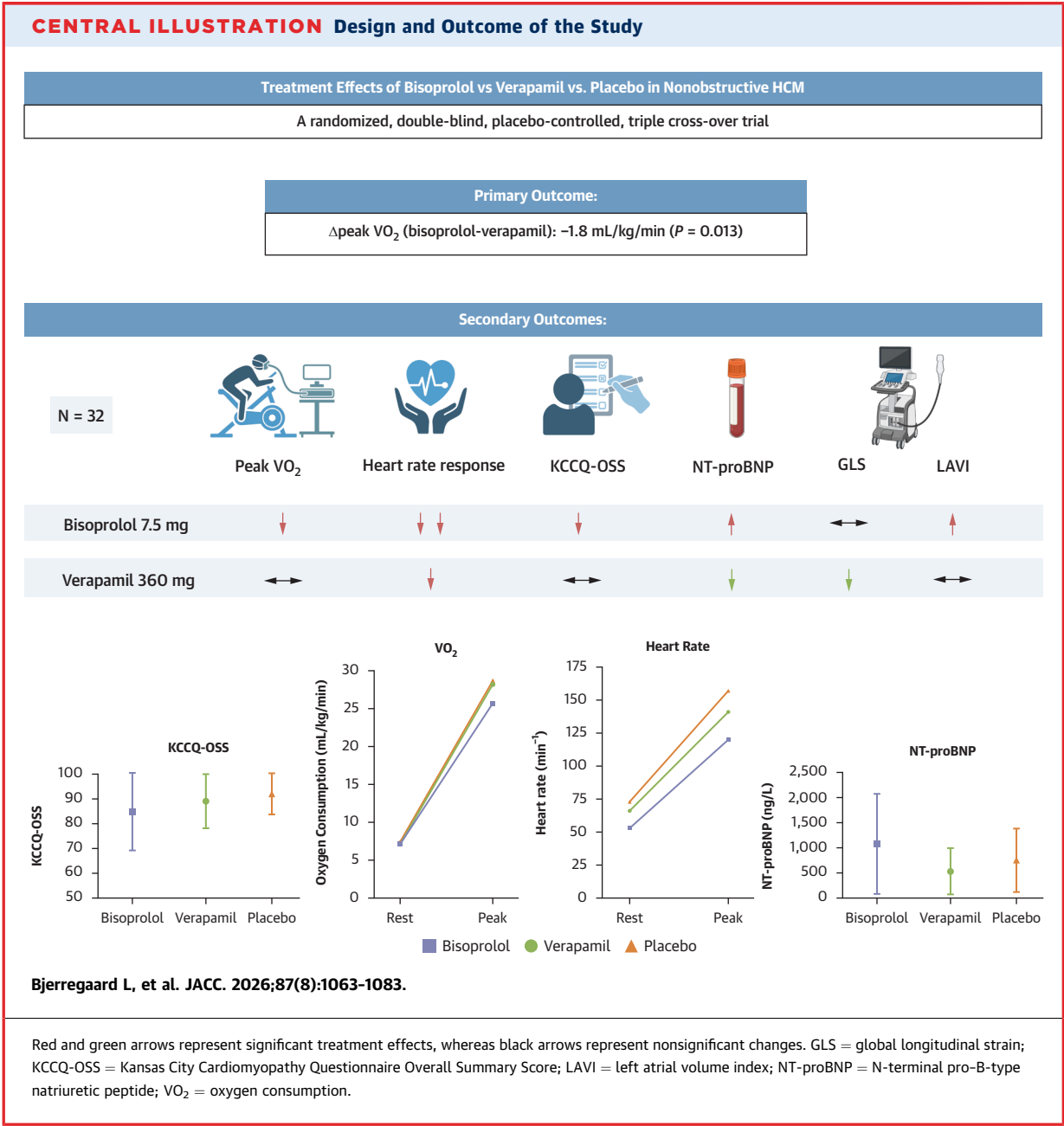
similarly to our findings.^{37–40} Palau et al³⁷ demonstrated that withdrawal of BB therapy in patients with HFpEF led to an increase in pVO₂ and improved quality of life. Moreover, their study identified a positive correlation between changes in pVO₂ and heart rate during peak exercise, underscoring the detrimental impact of BB’s negative chronotropic effect. In line with pVO₂ being an independent predictor of outcomes in HFpEF,^{41–44} pVO₂ has also been shown to independently predict disease progression and mortality in HCM.^{35,36,45,46} Similarly, an increased VE/VCO₂ slope is associated with increased risk of heart failure, death, sudden cardiac death, and transplantation in HCM,^{35,36,45,46} and there is a close relationship between pulmonary congestion, pulmonary vascular resistance, and VE/VCO₂ slope.⁴⁷ A recent study by Gudmundsdottir et al⁴⁸ demonstrated an improvement in pVO₂ and VE/VCO₂ slope after 12 weeks of exercise training in patients with nonobstructive HCM. In contrast, we demonstrated that bisoprolol and verapamil did not improve pVO₂ or affect the VE/VCO₂ slope. Likewise, we found no effect of bisoprolol or verapamil on AT. These differences in effect of exercise training and drug therapy may arise from different targets for the interventions, whereas exercise training is well known to improve the oxidative metabolism, not only in the heart, but throughout the body and particularly in skeletal muscle.^{49–51}

Bisoprolol and verapamil was chosen as intervention in the present trial for their high degree of cardiospecificity. Therefore, an effect on peripheral oxidative metabolism or ventilatory efficiency should not be expected. The unchanged AT and VE/VCO₂ slope in the present study may indicate that bisoprolol and verapamil did not affect the peripheral oxidative and regulative metabolisms or cause pulmonary congestion.

LV VOLUMES AND FUNCTION. The negative chronotropic effects of BB and CCB prolongs the diastolic phase, which is thought to enhance LV filling.^{16,52–54} Findings from our study substantiate that this effect is present during bisoprolol treatment, as LV end-systolic and end-diastolic volumes increased at rest. An increase in stroke volume (+5 mL), however, did not reach statistical significance although indicated by the resting O₂-pulse increased with bisoprolol. At peak exercise, O₂-pulse significantly increased with both bisoprolol and verapamil, but echocardiographic confirmation of increase in LV volumes was not performed during exercise. Similar observations have been reported in patients with heart failure with reduced ejection fraction, where BB therapy reduced heart rate and increased O₂-pulse, but without improving pVO₂.⁵⁵ In addition, LV ejection fraction is often normal or supranormal in HCM despite significant systolic dysfunction.⁵⁶ Neither bisoprolol nor verapamil changed the ejection fraction, although verapamil did improve LV GLS, which suggests a beneficial effect of verapamil treatment on LV systolic function.

The expected negative chronotropic effect of bisoprolol and verapamil contributed to the reduced RPP as an estimate of myocardial oxygen demand and work was confirmed in this study. These changes were present not only at rest, but also at peak exercise, though they were only borderline significant with verapamil at peak exercise. However, the general assumption that reduction in myocardial oxygen demand during BB and CCB treatment would translate into improved functional capacity and reduction in symptoms was not confirmed in this study.

Apart from the overt negative chronotropic effects of bisoprolol and verapamil assessed by heart rate, a surrogate marker of the negative inotropic effects on contraction and relaxation was assessed by diastolic parameters. The increased LV end-diastolic volume illustrated with bisoprolol treatment could reflect either increased LV filling pressure or improved relaxation of the LV. The higher LAVI, NT-proBNP, and TR pressure gradient during bisoprolol treatments suggest an increase in the LV filling pressure rather than improved relaxation, and may cause symptoms of backward failure.^{34,57} These are important clinical findings that may relate to the impaired functional capacity with bisoprolol treatment, as supported by the study by Le et al,⁵⁸ who demonstrated that LAVI correlates with pVO₂ in HCM. Similar effect of increased LAVI and the TR pressure gradient did not reach statistical significance during verapamil treatment. On the contrary, the increased e’ velocities, and the reduced NT-



proBNP with verapamil, may suggest improved relaxation and reduced LV filling pressure. Consequently, the overall effect of verapamil on diastolic parameters were more ambiguous. One possible explanation for these findings may be distinct differences in the intracellular signaling mechanisms of BB and CCB.⁵⁹⁻⁶²

SYMPTOMS. The NYHA classification was unchanged by the treatments, but may also be less sensitive for detecting moderate changes. Bisoprolol was

associated with a 6.6-points lower KCCQ-OSS compared with placebo. The KCCQ-OSS represents patients' physical function, symptoms, and quality of life.⁶³⁻⁶⁶ A 5-point decrease in KCCQ-OSS is associated with a 10% relative risk of cardiovascular death and a 2.5 mL/kg/min reduction in pVO_2 in patients with heart failure with reduced ejection fraction.⁶⁷ We observed a 6.6 points reduction in KCCQ-OSS accompanied by a 2.5 mL/kg/min decrease in pVO_2 . This relationship between KCCQ-OSS scores and

pVO₂ appears to be more pronounced in obstructive HCM. In EXPLORER-HCM,²⁰ a 9.1-point increase in KCCQ-OSS was associated with a 1.4-mL/kg/min improvement in pVO₂, and in SEQUOIA-HCM,⁶⁸ a 7-point increase was associated with a 1.8-mL/kg/min improvement. These findings suggest that in non-obstructive HCM, the relationship between symptoms and functional capacity aligns more closely with that observed in patients with heart failure with reduced and preserved ejection fractions, respectively, rather than with obstructive HCM patients.

STUDY COHORT. This study intended to include patients with significant myocardial disease without LVOT obstruction as excluded by extensive testing. As illustrated by the baseline characteristics, these patients did have signs of significant myocardial disease with severe hypertrophy, increased NT-proBNP levels, and severely enlarged left atria. Yet their functional capacity was only mildly affected, as well as their symptoms including a number of asymptomatic patients.

Nonobstructive HCM is traditionally defined by a resting gradient <30 mm Hg or a provoked gradient <50 mm Hg.^{4,5,49,50} We adopted a stricter definition to minimize the risk of borderline gradients leading to obstruction during the trial. Given the variety of stimuli that can induce gradients, alternative methods could have been applied. However, we also evaluated each patient's history of obstruction to reduce diagnostic uncertainty and ensure the inclusion of an exclusively nonobstructive HCM cohort.

STUDY DRUGS, DOSAGES, AND DURATION. In this study, bisoprolol was selected based on its high β_1 -selectivity, long duration of action, moderate lipophilicity, and stable volume of distribution, all of which made it a preferable option among β_1 -selective BBs for this trial.⁶⁹ Verapamil is the most frequently used nondihydropyridine CCB in HCM. This trial used daily doses of 7.5 mg bisoprolol and 360 mg verapamil, corresponding to 75% of maximal recommended daily dosage (10 mg for bisoprolol and 480 mg for verapamil).^{70,71} This approach was intended to reflect current clinical practice, where doses are titrated to clinical effect or maximum tolerated dose. The maximal recommended doses were avoided because we anticipated increased risk of side-effects. Both drugs are receptor antagonists with a relatively fast onset of effect and half-lives of 9 to 12 hours (bisoprolol) and 3 to 7 hours (verapamil), ensuring that steady state could be reached within a few days. Likewise, the short elimination time ensured complete washout between treatment periods. As such, the trial was designed to avoid

carryover effects in the triple cross-over design, which was confirmed by statistical testing. In comparison, cardiac myosin inhibitors have longer half-lives (mavacamten: 6-9 days, steady state after 6 weeks; aficamten: 75-85 hours, steady state after 10-12 days^{72,73}), which is why a similar cross-over trial is not feasible in cardiac myosin inhibitors.

STUDY LIMITATIONS. This cohort represents a non-obstructive HCM population in which patients have signs of severe myocardial disease, though presenting with fewer symptoms compared with the study populations of recent larger trials of cardiac myosin inhibitor in both obstructive and nonobstructive HCM. A notable number of patients were classified as NYHA I but fulfilled other criteria for inclusion, such as elevated NT-proBNP or nonsustained VT. This may limit the ability to detect changes in measures of feel and function, as well as the applicability of findings to more symptomatic patients. Nevertheless, 37% of these patients were treated with BBs or CCBs before enrollment. This study provides data on what to expect when treating patients with nonobstructive HCM. Evaluation of treatment effects in these patients are relevant for a large number of patients with nonobstructive HCM worldwide.

A comprehensive assessment of diastolic function would include right heart catheterization, including measurement of pulmonary capillary wedge pressure, which was not performed in this study. Owing to the overall complexity of the trial, other modalities were prioritized. The echocardiographic measures should be interpreted as surrogate markers of diastolic function.

The pronounced negative chronotropic effects of bisoprolol and verapamil potentially threaten integrity of the blinding, but difference in potency between the 2 treatments was unknown before the study was initiated. Therefore, a reduction in heart rate did not disclose the active treatments. Conversely, blinding integrity could potentially be affected for the placebo treatment. To compensate for this, KCCQ-OSS and NYHA classification were always performed before obtaining heart rate, blood pressure, electrocardiography, echocardiography, and CPET in all patients. Thus, treatment allocation remained unknown to investigators and patients at the time of evaluation. Drug-specific side-effects are an inherent threat to blinding integrity in all blinded studies, including this one. There are a number of overlapping side-effects between bisoprolol and verapamil that minimize this risk, but a few differences in very specific side-effects exist.

This trial reflects only short-term effects of bisoprolol and verapamil. We acknowledge that long-

term treatment with bisoprolol and verapamil may lead to receptor up-regulation and other physiologic adaptations not captured in this study.

PERSPECTIVES. Recognizing the hemodynamic differences between obstructive and nonobstructive HCM is crucial for effective management of symptoms, risks, and complications in a heterogeneous HCM population.^{74,75} The findings of the present study in nonobstructive HCM patients, underscores that the hemodynamic and symptomatic benefits of BB observed in obstructive HCM patients²¹ cannot be directly extrapolated to nonobstructive HCM patients. This highlights the need for tailored, phenotype-specific therapeutic approaches.^{76,77} This is particularly relevant in the context of emerging cardiac myosin inhibitors, in which improvement of functional capacity, symptom burden, and biomarkers have been demonstrated in obstructive HCM.^{20,78} Although potential benefits of such treatment in nonobstructive HCM was suggested in the MAVERICK (mavacamten)⁷⁹ and REDWOOD (aficamten)⁸⁰ phase 2 trials, the clinical efficacy was not confirmed in the phase 3 trials of ODYSSEY-HCM.⁸¹ We are still awaiting results for aficamten treatment in nonobstructive HCM from ACACIA-HCM (Phase 3 Trial to Evaluate the Efficacy and Safety of Aficamten Compared to Placebo in Adults With Symptomatic nHCM; [NCT06081894](#)).

BB treatment may be an important confounder in these trials. In ODYSSEY-HCM, 78% of patients were treated with BBs during the trial.⁸¹ Based on results from the present study, BBs may have affected the primary outcome in ODYSSEY-HCM and may have blurred the true effect of mavacamten in non-obstructive HCM. In an attempt to exclude the effects of BBs, the MAPLE-HCM trial compared metoprolol and aficamten head-to-head with no placebo arm.⁸² That trial showed an increase in pVO₂ and reduction in LVOT gradient with aficamten and a reduction in pVO₂ together with unchanged LVOT gradients with metoprolol. MAPLE-HCM also illustrated an increase in NT-proBNP and left atrial dimension, similarly to

the present study. This may suggest that the negative chronotropic effect of metoprolol may also negatively affect the pVO₂ in obstructive HCM. This may particularly be the case in patients in whom metoprolol fails to reduce the outflow tract obstruction.

In summary, despite historical clinical trials and contemporary attempts to validate treatment effects of BBs, CCBs, and myosin inhibitors in non-obstructive HCM, there are still no documented treatment options for this patient category.

CONCLUSIONS

The negative chronotropic effects of bisoprolol and verapamil were confirmed in this study. Despite the expected remodeling associated with a reduced heart rate, the marked heart rate reduction with bisoprolol treatment led to a decrease in pVO₂. In contrast, the pVO₂ remained unchanged with verapamil treatment, even with a moderate reduction in heart rate. This difference may be explained by their distinct pharmacodynamic effects on the myocardium, resulting in different impacts on LV function. Verapamil improved diastolic myocardial velocities, GLS, and NT-proBNP levels, indicating enhanced myocardial performance, whereas bisoprolol did not improve these parameters. This study provides new insights into the mechanisms of these drugs that might be taken into consideration in management of nonobstructive HCM.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

Funding support was provided by the Novo Nordisk Foundation (NNF21OC071881) and the Danish Regions-Medicine and Treatment Fund. Dr Dybro and Dr Hvitfeldt Poulsen have served on the steering committee for the MAPLE-HCM trial sponsored by Cytokinetics. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Dr Louise Bjerregaard, Department of Cardiology, Aarhus University Hospital, Palle Juul-Jensens Bld. 99, 8200 Aarhus N, Denmark. E-mail: louise.bjerregaard1202@gmail.com.

REFERENCES

1. Maron BJ, Rowin EJ, Udelson JE, Maron MS. Clinical spectrum and management of heart failure in hypertrophic cardiomyopathy. *JACC Heart Fail.* 2018;6(5):353-363.
2. Houston BA, Stevens GR. Hypertrophic cardiomyopathy: a review. *Clin Med Insights Cardiol.* 2014;8(Suppl 1):53-65.
3. Sherrid MV, Barac I, McKenna WJ, et al. Multicenter study of the efficacy and safety of disopyramide in obstructive hypertrophic cardiomyopathy. *J Am Coll Cardiol.* 2005;45(8):1251-1258.
4. Arbelo E, Protonotarios A, Gimeno JR, et al. 2023 ESC guidelines for the management of cardiomyopathies. *Eur Heart J.* 2023;44(37):3503-3626.
5. Writing Committee Members, Ommen SR, Ho CY, Asif IM, et al. 2024 AHA/ACC/AMSSM/HRS/PACES/SCMR guideline for the management of hypertrophic cardiomyopathy: a report of the American Heart Association/American College of Cardiology Joint Committee on Clinical Practice Guidelines. *J Am Coll Cardiol.* 2024;83(23):2324-2405.
6. Bonow RO, Frederick TM, Bacharach SL, et al. Atrial systole and left ventricular filling in hypertrophic cardiomyopathy: effect of verapamil. *Am J Cardiol.* 1983;51(8):1386-1391.
7. Bonow RO, Ostrow HG, Rosing DR, et al. Effects of verapamil on left ventricular systolic and

- diastolic function in patients with hypertrophic cardiomyopathy: pressure-volume analysis with a nonimaging scintillation probe. *Circulation*. 1983;68(5):1062–1073.
8. Bonow RO, Dilsizian V, Rosing DR, Maron BJ, Bacharach SL, Green MV. Verapamil-induced improvement in left ventricular diastolic filling and increased exercise tolerance in patients with hypertrophic cardiomyopathy: short- and long-term effects. *Circulation*. 1985;72(4):853–864.
 9. Cohen LS, Braunwald E. Amelioration of angina pectoris in idiopathic hypertrophic subaortic stenosis with beta-adrenergic blockade. *Circulation*. 1967;35(5):847–851.
 10. Flamm MD, Harrison DC, Hancock EW. Muscular subaortic stenosis. Prevention of outflow obstruction with propranolol. *Circulation*. 1968;38(5):846–858.
 11. Adelman AG, Shah PM, Gramiak R, Wigle ED. Long-term propranolol therapy in muscular subaortic stenosis. *Br Heart J*. 1970;32(6):804–811.
 12. Stenson RE, Flamm MD Jr, Harrison DC, Hancock EW. Hypertrophic subaortic stenosis. Clinical and hemodynamic effects of long-term propranolol therapy. *Am J Cardiol*. 1973;31(6):763–773.
 13. Ostman-Smith I, Wettrell G, Riesenfeld T. A cohort study of childhood hypertrophic cardiomyopathy: improved survival following high-dose beta-adrenoceptor antagonist treatment. *J Am Coll Cardiol*. 1999;34(6):1813–1822.
 14. Maron BJ. Hypertrophic cardiomyopathy: a systematic review. *JAMA*. 2002;287.
 15. Rosing DR, Kent KM, Borer JS, Seides SF, Maron BJ, Epstein SE. Verapamil therapy: a new approach to the pharmacologic treatment of hypertrophic cardiomyopathy. I. Hemodynamic effects. *Circulation*. 1979;60(6):1201–1207.
 16. Bonow RO, Rosing DR, Bacharach SL, et al. Effects of verapamil on left ventricular systolic function and diastolic filling in patients with hypertrophic cardiomyopathy. *Circulation*. 1981;64(4):787–796.
 17. Pollick C, Kimball B, Henderson M, Wigle ED. Disopyramide in hypertrophic cardiomyopathy. I. Hemodynamic assessment after intravenous administration. *Am J Cardiol*. 1988;62(17):1248–1251.
 18. Gilligan DM, Chan WL, Joshi J, et al. A double-blind, placebo-controlled crossover trial of nadolol and verapamil in mild and moderately symptomatic hypertrophic cardiomyopathy. *J Am Coll Cardiol*. 1993;21(7):1672–1679.
 19. Gistri R, Cecchi F, Choudhury L, et al. Effect of verapamil on absolute myocardial blood flow in hypertrophic cardiomyopathy. *Am J Cardiol*. 1994;74(4):363–368.
 20. Olivetto I, Oreziak A, Barriales-Villa R, et al. Mavacamten for treatment of symptomatic obstructive hypertrophic cardiomyopathy (EXPLORER-HCM): a randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet*. 2020;396(10253):759–769.
 21. Dybro AM, Rasmussen TB, Nielsen RR, Andersen MJ, Jensen MK, Poulsen SH. Randomized trial of metoprolol in patients with obstructive hypertrophic cardiomyopathy. *J Am Coll Cardiol*. 2021;78(25):2505–2517.
 22. Fletcher GF, Balady G, Froelicher VF, Hartley LH, Haskell WL, Pollock ML. Exercise standards. A statement for healthcare professionals from the American Heart Association. *Circulation*. 1995;91(2):580–615.
 23. Gobel F, Nordstrom L, Nelson R, Jorgensen C, Wang Y. The rate-pressure product as an index of myocardial oxygen consumption during exercise in patients with angina pectoris. *Circulation*. 1978;57(3):549–556.
 24. Kitamura K, Jorgensen CR, Gobel FL, Taylor HL, Wang Y. Hemodynamic correlates of myocardial oxygen consumption during upright exercise. *J Appl Physiol*. 1972;32(4):516–522.
 25. Nelson R, Gobel F, Jorgensen C, Wang K, Wang Y, Taylor H. Hemodynamic predictors of myocardial oxygen consumption during static and dynamic exercise. *Circulation*. 1974;50(6).
 26. Mezzani A. Cardiopulmonary exercise testing: basics of methodology and measurements. *Ann Am Thorac Soc*. 2017;14(Suppl 1):S3–S11.
 27. Leclerc K. Cardiopulmonary exercise testing: a contemporary and versatile clinical tool. *Cleve Clin J Med*. 2017;84(2):161–168.
 28. Nagueh SF, Smiseth OA, Appleton CP, et al. Recommendations for the evaluation of left ventricular diastolic function by echocardiography: an update from the American Society of Echocardiography and the European Association of Cardiovascular Imaging. *J Am Soc Echocardiogr*. 2016;29(4):277–314.
 29. Nagueh SF. Left ventricular diastolic function: understanding pathophysiology, diagnosis, and prognosis with echocardiography. *JACC Cardiovasc imaging*. 2020;13(1).
 30. Finocchiaro G, Haddad F, Pavlovic A, et al. How does morphology impact on diastolic function in hypertrophic cardiomyopathy? A single centre experience. *BMJ Open*. 2014;4(6):e004814.
 31. Saccheri MC, Cianciulli TF, Lax JA, et al. Impaired myocardial function in hypertrophic cardiomyopathy. *Echocardiography*. 2009;26(6):657–664.
 32. Kalra A, Harris KM, Maron BA, et al. Relation of Doppler tissue imaging parameters with heart failure progression in hypertrophic cardiomyopathy. *Am J Cardiol*. 2016;117(11):1808–1814.
 33. Williams LK, Frenneaux MP, Steeds RP. Echocardiography in hypertrophic cardiomyopathy diagnosis, prognosis, and role in management. *Eur J Echocardiogr*. 2009;10(8):iii9–iii14.
 34. Geske JB, Sorajja P, Nishimura RA, Ommen SR. The relationship of left atrial volume and left atrial pressure in patients with hypertrophic cardiomyopathy: an echocardiographic and cardiac catheterization study. *J Am Soc Echocardiogr*. 2009;22(8):961–966.
 35. Sorajja P, Allison T, Hayes C, Nishimura RA, Lam CS, Ommen SR. Prognostic utility of metabolic exercise testing in minimally symptomatic patients with obstructive hypertrophic cardiomyopathy. *Am J Cardiol*. 2012;109(10):1494–1498.
 36. Coats CJ, Rantell K, Bartnik A, et al. Cardiopulmonary exercise testing and prognosis in hypertrophic cardiomyopathy. *Circ Heart Fail*. 2015;8(6):1022–1031.
 37. Palau P, Seller J, Domínguez E, et al. Effect of β -blocker withdrawal on functional capacity in heart failure and preserved ejection fraction. *J Am Coll Cardiol*. 2021;78(21):2042–2056.
 38. Wernhart S, Papathanasiou M, Mahabadi AA, Rassaf T, Luedike P. Betablockers reduce oxygen pulse increase and performance in heart failure patients with preserved ejection fraction. *Int J Cardiol*. 2023;370:309–318.
 39. Domínguez E, Palau P, Nunez E, et al. Heart rate response and functional capacity in patients with chronic heart failure with preserved ejection fraction. *ESC Heart Fail*. 2018;5(4):579–585.
 40. Yuasa N, Obokata M, Harada T, et al. Characterization and prognostic importance of chronotropic incompetence in heart failure with preserved ejection fraction. *J Cardiol*. 2024;83(2):113–120.
 41. Sato T, Yoshihisa A, Kanno Y, et al. Cardiopulmonary exercise testing as prognostic indicators: comparisons among heart failure patients with reduced, mid-range and preserved ejection fraction. *Eur J Prev Cardiol*. 2017;24(18):1979–1987.
 42. Dhakal BP, Malhotra R, Murphy RM, et al. Mechanisms of exercise intolerance in heart failure with preserved ejection fraction: the role of abnormal peripheral oxygen extraction. *Circ Heart Fail*. 2015;8(2):286–294.
 43. Malhotra R, Bakken K, d'Elia E, Lewis GD. Cardiopulmonary exercise testing in heart failure. *JACC Heart Fail*. 2016;4(8):607–616.
 44. Guazzi M, Bandera F, Ozemek C, Systrom D, Arena R. Cardiopulmonary exercise testing: what is its value? *J Am Coll Cardiol*. 2017;70(13):1618–1636.
 45. Magri D, Re F, Limongelli G, et al. Heart failure progression in hypertrophic cardiomyopathy—possible insights from cardiopulmonary exercise testing. *Circ J*. 2016;80(10):2204–2211.
 46. Magri D, Limongelli G, Re F, et al. Cardiopulmonary exercise test and sudden cardiac death risk in hypertrophic cardiomyopathy. *Heart*. 2016;102(8):602–609.
 47. Lewis GD, Shah RV, Pappagianopoulos PP, Systrom DM, Semigran MJ. Determinants of ventilatory efficiency in heart failure: the role of right ventricular performance and pulmonary vascular tone. *Circ Heart Fail*. 2008;1(4):227–233.
 48. Gudmundsdottir HL, Axelsson Raja A, Rossing K, et al. Exercise training in patients with hypertrophic cardiomyopathy without left ventricular outflow tract obstruction: a randomized clinical trial. *Circulation*. 2025;151(2):132–144.
 49. Porter C, Reidy PT, Bhattarai N, Sidossis LS, Rasmussen BB. Resistance exercise training alters mitochondrial function in human skeletal muscle. *Med Sci Sports Exerc*. 2015;47(9):1922–1931.
 50. Lundby C, Jacobs RA. Adaptations of skeletal muscle mitochondria to exercise training. *Exp Physiol*. 2016;101(1):17–22.

51. Martin TP. Predictable adaptations by skeletal muscle mitochondria to different exercise training workloads. *Comp Biochem Physiol B*. 1987;88(1):273-276.
52. Hanrath P, Mathey DG, Kremer P, Sonntag F, Bleifeld W. Effect of verapamil on left ventricular isovolumic relaxation time and regional left ventricular filling in hypertrophic cardiomyopathy. *Am J Cardiol*. 1980;45(6):1258-1264.
53. Zhu M, Reyes KRL, Bilgili G, et al. Medical therapies to improve left ventricular outflow obstruction and diastolic function in hypertrophic cardiomyopathy. *JACC Adv*. 2023;2(8):100622.
54. Fifer MA, Vlahakes GJ. Management of symptoms in hypertrophic cardiomyopathy. *Circulation*. 2008;117(3):429-439.
55. Gullestad L, Manhenke C, Aarsland T, et al. Effect of metoprolol CR/XL on exercise tolerance in chronic heart failure—a substudy to the MERIT-HF trial. *Eur J Heart Fail*. 2001;3(4):463-468.
56. Halliday BP, Senior R, Pennell DJ. Assessing left ventricular systolic function: from ejection fraction to strain analysis. *Eur Heart J*. 2021;42(7):789-797.
57. Nagueh SF. Noninvasive assessment of left ventricular filling pressure. *Eur J Heart Fail*. 2018;20(1):38-48.
58. Le V-V, Perez MV, Wheeler MT, Myers J, Schnitger I, Ashley EA. Mechanisms of exercise intolerance in patients with hypertrophic cardiomyopathy. *American Heart Journal*. 2009;158(3):e27-e34.
59. Rehsia NS, Dhalla NS. Mechanisms of the beneficial effects of beta-adrenoceptor antagonists in congestive heart failure. *Exp Clin Cardiol*. 2010;15(4):e86-e95.
60. Elliott WJ, Ram CV. Calcium channel blockers. *J Clin Hypertens (Greenwich)*. 2011;13(9):687-689.
61. Tucker WD, Sankar P, Theetha Kariyanna P. *Selective beta-1 blockers*. Treasure Island (FL); 2025.
62. Palandri C, Santini L, Argiro A, et al. Pharmacological management of hypertrophic cardiomyopathy: from bench to bedside. *Drugs*. 2022;82(8):889-912.
63. Green CP, Porter CB, Bresnahan DR, Spertus JA. Development and evaluation of the Kansas City Cardiomyopathy Questionnaire: a new health status measure for heart failure. *J Am Coll Cardiol*. 2000;35(5):1245-1255.
64. Spertus JA, Jones PG, Sandhu AT, Arnold SV. Interpreting the Kansas City Cardiomyopathy Questionnaire in clinical trials and clinical care: JACC state-of-the-art review. *J Am Coll Cardiol*. 2020;76(20):2379-2390.
65. Garcia RA, Benton MC, Spertus JA. Patient-reported outcomes in patients with cardiomyopathy. *Curr Cardiol Rep*. 2021;23(7):91.
66. Nassif M, Fine JT, Dolan C, et al. Validation of the Kansas City Cardiomyopathy Questionnaire in symptomatic obstructive hypertrophic cardiomyopathy. *JACC Heart Fail*. 2022;10(8):531-539.
67. Flynn KE, Lin L, Moe GW, et al. Relationships between changes in patient-reported health status and functional capacity in outpatients with heart failure. *Am Heart J*. 2012;163(1):88-94.
68. Maron MS, Masri A, Nassif ME, et al. Aficamten for symptomatic obstructive hypertrophic cardiomyopathy. *N Engl J Med*. 2024;390(20):1849-1861.
69. Roston TM, Chua D, Lum E, Krahn AD. Switching between β -blockers: an empiric tool for the cardiovascular practitioner. *Can J Cardiol*. 2019;35(4):539-543.
70. Danish Medicines Agency. Bisoprolol “Krka.” <https://pro.medicin.dk/Medicin/Praeparater/7378>
71. Danish Medicines Agency. Isoptin Retard. <https://pro.medicin.dk/Medicin/Praeparater/233>
72. Grillo MP, Erve JCL, Dick R, et al. In vitro and in vivo pharmacokinetic characterization of mavacamten, a first-in-class small molecule allosteric modulator of beta cardiac myosin. *Xenobiotica*. 2019;49(6):718-733.
73. Hartman JJ, Hwee DT, Robert-Paganin J, et al. Aficamten is a small-molecule cardiac myosin inhibitor designed to treat hypertrophic cardiomyopathy. *Nat Cardiovasc Res*. 2024;3(8):1003-1016.
74. Maron BJ, Ommen SR, Semsarian C, Spirito P, Olivetto I, Maron MS. Hypertrophic cardiomyopathy: present and future, with translation into contemporary cardiovascular medicine. *J Am Coll Cardiol*. 2014;64(1):83-99.
75. Maron BJ, Maron MS. The 20 advances that have defined contemporary hypertrophic cardiomyopathy. *Trends Cardiovasc Med*. 2015;25(1):54-64.
76. Maron BJ, Desai MY, Nishimura RA, et al. Management of hypertrophic cardiomyopathy: JACC state-of-the-art review. *J Am Coll Cardiol*. 2022;79(4):390-414.
77. Olalekan SO, Bakare OO, Okwute PG, Osonuga IO, Adeyanju MM, Edema VB. Hypertrophic cardiomyopathy: insights into pathophysiology and novel therapeutic strategies from clinical studies. *Egypt Heart J*. 2025;77(1):5.
78. Coats CJ, Maron MS, Abraham TP, et al. Exercise capacity in patients with obstructive hypertrophic cardiomyopathy: SEQUOIA-HCM baseline characteristics and study design. *JACC Heart Fail*. 2024;12(1):199-215.
79. Ho CY, Mealiffe ME, Bach RG, et al. Evaluation of mavacamten in symptomatic patients with nonobstructive hypertrophic cardiomyopathy. *J Am Coll Cardiol*. 2020;75(21):2649-2660.
80. Masri A, Sherrid MV, Abraham TP, et al. Efficacy and safety of aficamten in symptomatic nonobstructive hypertrophic cardiomyopathy: results from the REDWOOD-HCM trial, cohort 4. *J Card Fail*. 2024;30(11):1439-1448.
81. Desai MY, Owens AT, Abraham T, et al. Mavacamten in symptomatic nonobstructive hypertrophic cardiomyopathy. *N Engl J Med*. 2025;393(10):961-972.
82. Garcia-Pavia P, Maron MS, Masri A, et al. Aficamten or metoprolol monotherapy for obstructive hypertrophic cardiomyopathy. *N Engl J Med*. 2025;393(10):949-960.

KEY WORDS beta-blocker, biomarker, calcium-antagonist, functional capacity, hypertrophic cardiomyopathy, myocardial function, symptoms

APPENDIX For supplemental material, please see the online version of this paper.



Go to <https://www.acc.org/jacc-journals-cme> to take the CME/MOC quiz for this paper.